

GE Healthcare

Signa HDx 16 Channel to 32 Channel System Upgrade

Technical Publication



5166011
Rev 8

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

Revision History

Date	Rev	Change Description
4-24-2006	1	Initial Release
7-13-2006	2	Section 2: Revised upgrade procedure information. Section 5: Corrected section reference. Corrected Run numbers in Sec. 5-2-5. Added Note to cable map in Sec. 5-4. Added note to discard lockwashers in Sec. 5-4-2 (MRIhc16256).
10-20-2006	3	Section 2: Updated flowchart to remove Infiniband Switch reference. Section 6: Design change required removal of Infiniband Switch installation. Updated cable maps and cabinet upgrade instructions.
5-11-2007	4	Section 5: Added WideOpen rear pedestal cover removal and installation procedures. Added note in Section 5-4-3 for installing Y-cables in rear pedestal (CQA 13100261). Section 6: Added System Cabinet references where applicable (CQA 13100285). Added Run 1351 fiber optic cable connection procedure for RFS/System Cabinet (CQA 13100283).
10-11-2007	5	Section 1: Added block containing M3335A to Upgrade catalogs flow chart in Section 1-9-2. Section 6: Added ground wire, Run 1349, to cable maps and also added the connection procedures to the 32 Channel cabinet and the PDU. (PQR 13100297)
9-18-2008	6	Added HDxt (Release 15.x) references in manual. Section 1: Updated system configuration with new ICN information. Section 2: Updated flowcharts with new ICN information. Section 6: Revised entire section to reflect the changes in installation for the new ICN modules.
22-Jun-2009	7	Added Section 6-6, <i>RFS or System Cabinet Upgrade for 1.5T and 3.0T with CAM Chassis</i> (PQR 13251206)
7-Oct-2010	8	Section 1: Updated system configuration by designating M3335BH catalog offering (ECR 2098339, PCN 290441). Section 6: Updated ICN upgrade procedures for installation of new 4170 ICN, part number 5337894-4 (ECO 2090162).

List of Effective Pages

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page	8	14	1	23 to 24	1	37	4	41 to 43	6
1 to 4	1	15	2	25	2	38 to 39	1	44 to 49	7
5 to 7	8	16	1	26 to 33	1	40	4	50 to 52	8
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13	5	22	4	36	2				

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SECTION 1 – OVERVIEW

1-1 UPGRADE INTRODUCTION

This publication documents the mechanical installation information for upgrading a Signa® HDx (Release 14.x) or HDxt (Release 15.x) 1.5T or 3.0T from 16 Channels to 32 Channels. Refer to illustrations 1–2 and 1–4 for pictorial overview of areas of system upgraded.

This Direction covers the following catalogs:

M3335PW – Signa HDx 3.0T 16 Channel to 32 Channel Upgrade

M3335PY – Signa HDx 1.5T 16 Channel to 32 Channel Upgrade

This upgrade adds:

- A second 16 Channel Switch, IRF Module, and associated cables
- The C–Bezel and harness to the LPCA

This upgrade may also require additional Catalogs to be installed. Refer to flowcharts on pages 11 (1.5T) and 13 (3.0T) for optional upgrade equipment that would be needed.



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1-2 SITE READY CHECK

Pre–installation work must be completed before equipment is delivered to avoid delays and confusion.

1-3 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The “Shipping Document” lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short. Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of the box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with “Technical Publication” boxes for a list of delivered documents.

1-4 UPGRADE SCHEDULE CONSIDERATIONS

FMIIs, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1-5 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period. File a report with:

- Call 1-800-548-3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1-6 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-6-1 MATERIAL POLICY OVERVIEW

1-6-1-1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1-6-1-2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-6-2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment de-installed for the HDx upgrade to GoldSeal.

1-6-3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new components were shipped to site for return of above material.
- Packing material

1-6-4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for any deinstallation procedures of the site.

1-6-5 REGION SPECIFIC RETURN INFORMATION

1-6-5-1 Sites in Americas

A GoldSeal return pre-printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1-877-251-4468 or 262-513-4147

For non-GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)-747-6997 or Dial Comm 8*579-6997
Fax: (414)-747-6855
Field ASPEN: 1-800-525-1516, Box #26041

1-6-5-2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS-Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1-6-5-3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS-E – GOLD SEAL OPERATION & RECYCLING CENTER
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL
Quai 45 à 49 – 91090 LISSES – FRANC
Tel: (+33) 1-60-91-57-35 or (+33) 1 64 97 55 41
Fax: (+33) 1-60-86-12-31

For non-Gold Seal returns in Europe:

European Renewable Resources Center
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2
Quai 45 à 49 – 91090 LISSES – FRANCE
TEL = (+33)-1-64-97-55-41 FAX = (+33)-1-60-86-12-31

1-6-6 PRODUCT LOCATOR INFORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1-8.

1-7 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at: <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: [Product Locator Support Central Page](http://supportcentral.ge.com/15563) – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-8 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Confirm back-up site information. Shut site down.
- Remove any obsolete equipment and cables.
- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1-9 UPGRADE SYSTEM CONFIGURATION

1-9-1 HDx 1.5T UPGRADE

Illustration 1-1 shows the Signa HDx 1.5T (Release 14.x) or HDxt (Release 15.x) 32 channel upgrade catalogs. Illustration 1-2 shows the equipment modified by the upgrade catalogs.

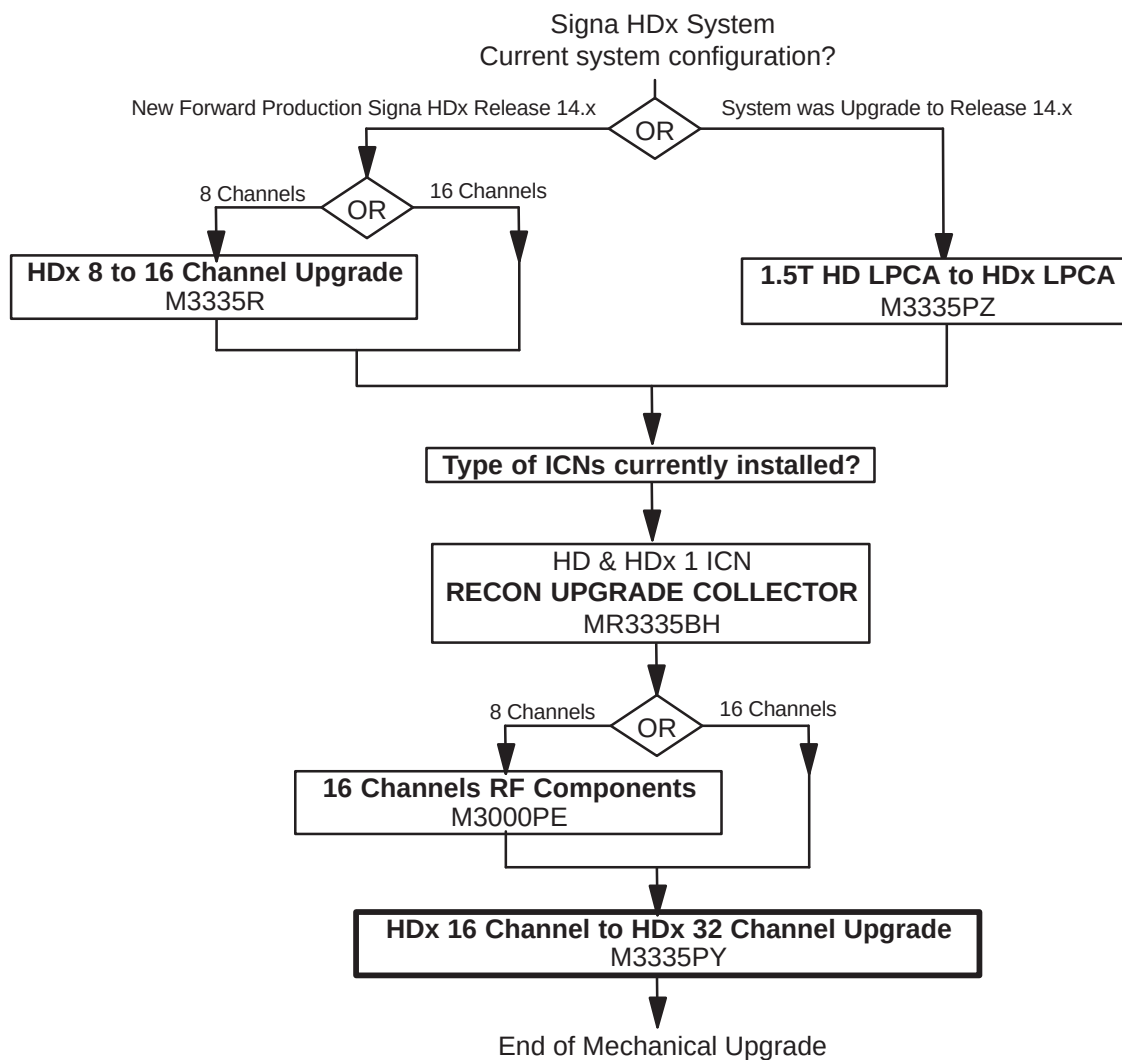


ILLUSTRATION 1-1
1.5T 32 Channels Upgrade Catalogs

1-9-1 HDx 1.5T UPGRADE (Continued)

Note: Only affected equipment shown.

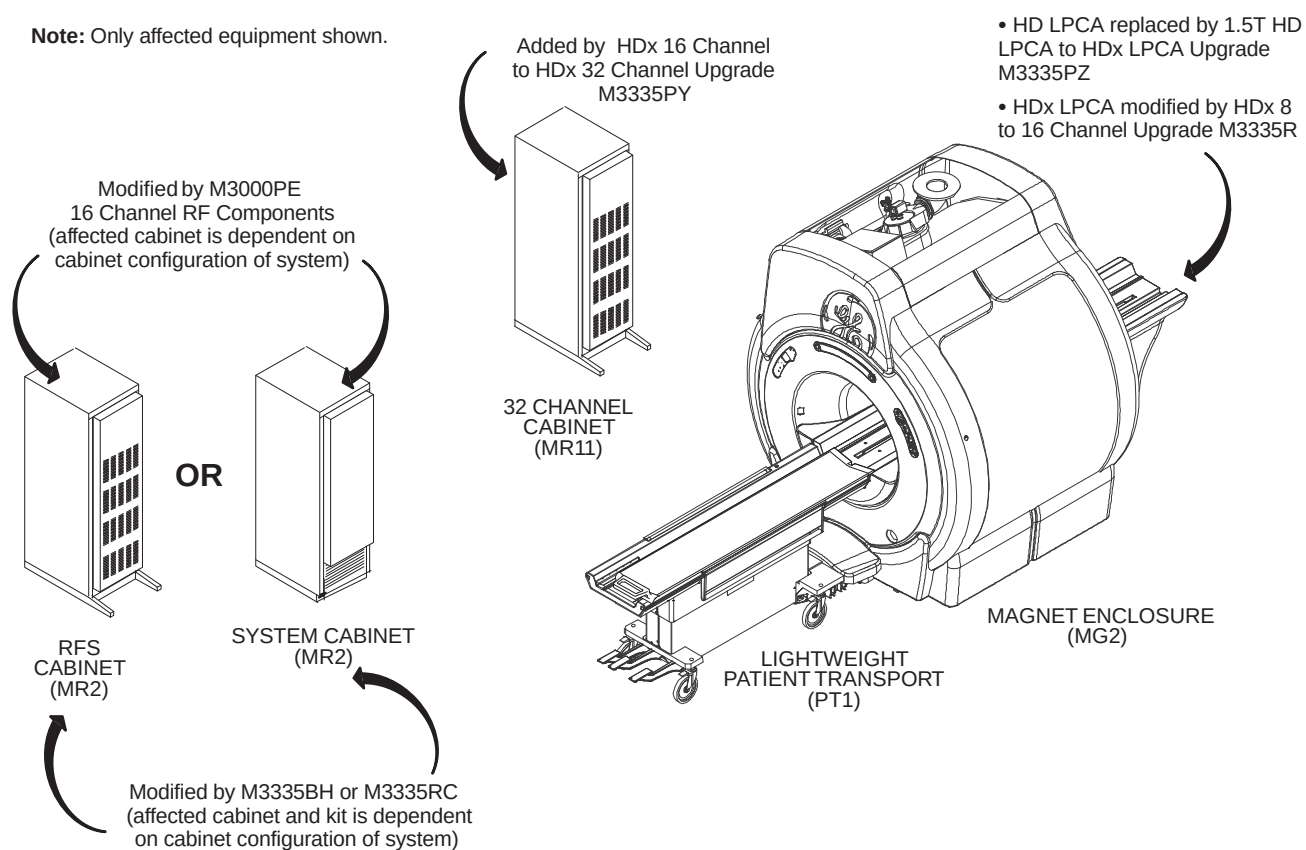


ILLUSTRATION 1-2
1.5T 32 Channel Upgrade Modified Equipment

1-9-2 HDx 3.0T UPGRADE

Illustration 1-3 shows the Signa HDx 3.0T (Release 14.x) or HDxt (Release 15.x) 32 channel upgrade catalogs. Illustration 1-4 shows the equipment modified by the upgrade catalogs.

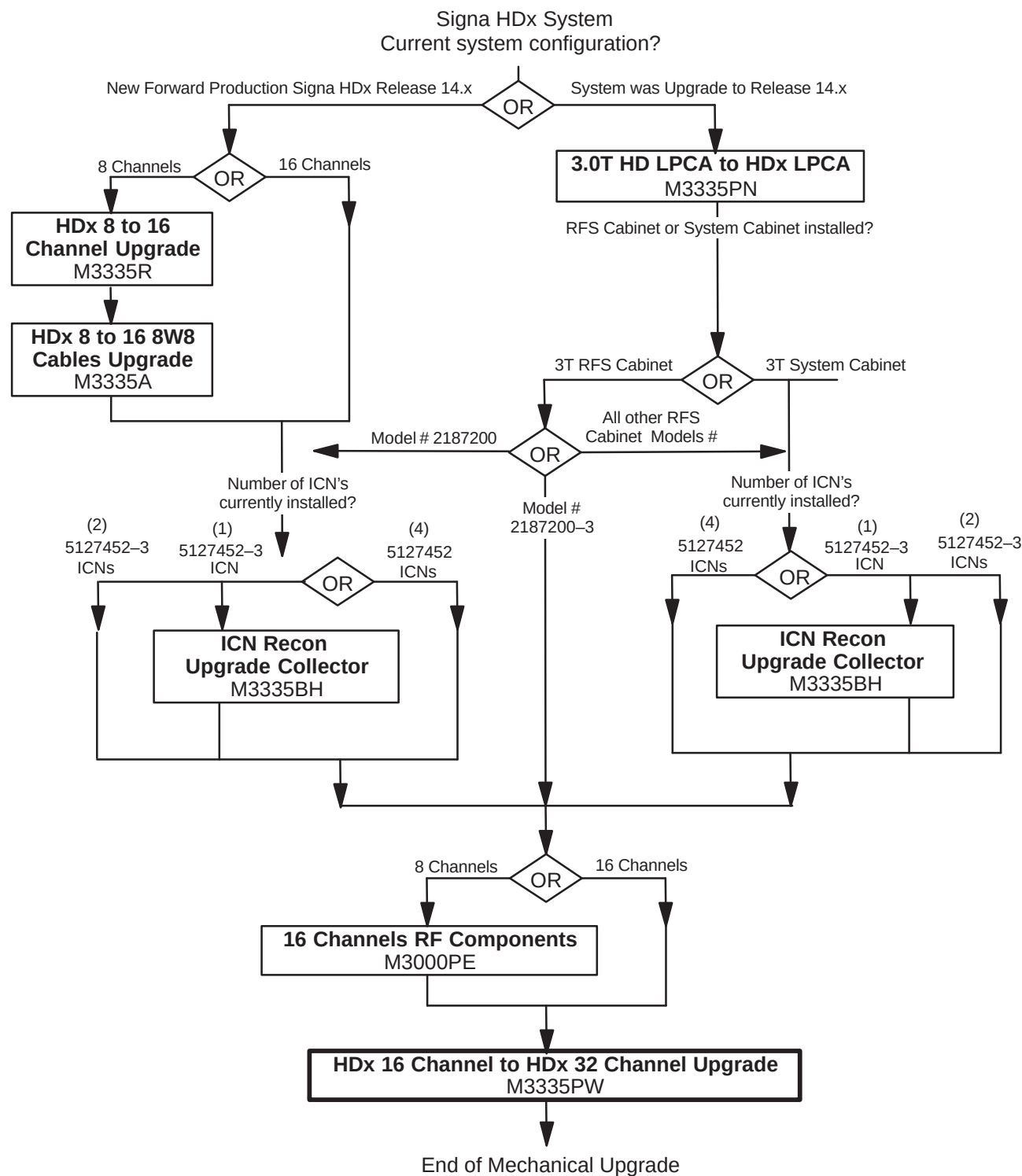


ILLUSTRATION 1-3
3.0T 32 Channels Upgrade Catalogs

1-9-2 HDx 3.0T UPGRADE (Continued)

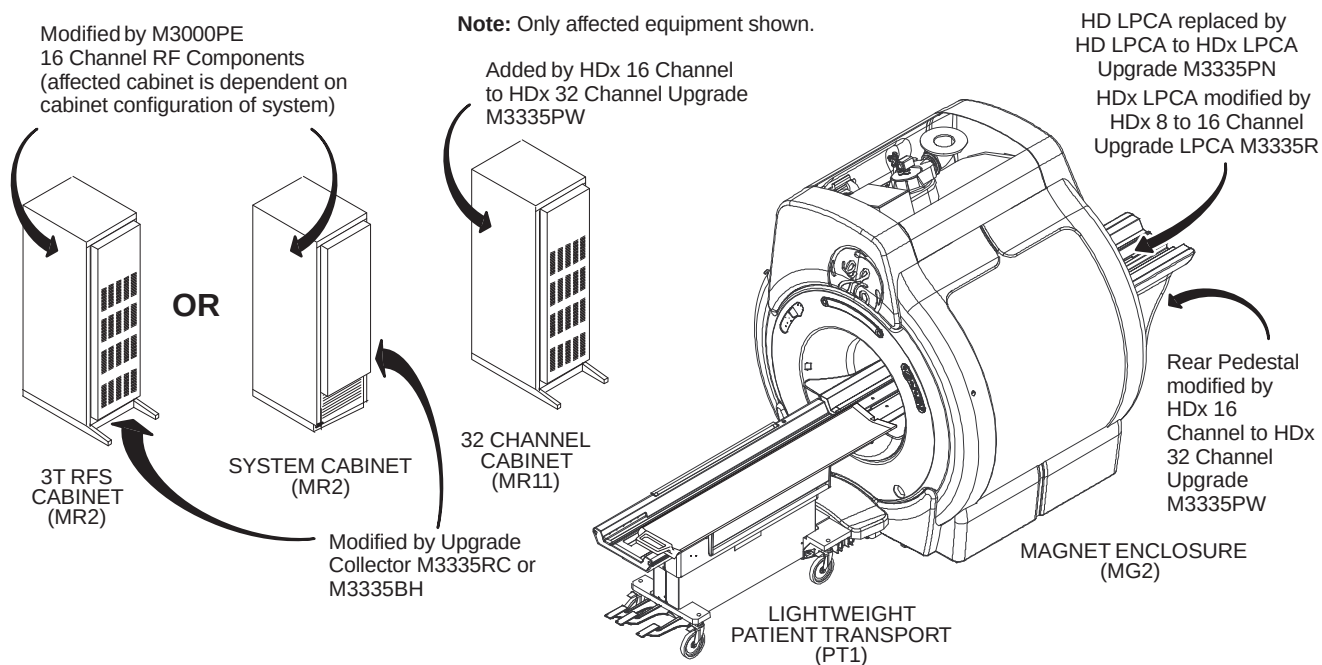


ILLUSTRATION 1-4
3.0T 32 Channel Upgrade Modified Equipment

SECTION 2 – UPGRADE INSTALLATION PROCESS

2-1 SITE READINESS

Any and all on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

2-2 UPGRADE PROCEDURE

The following upgrade flow chart contains information for upgrading a Signa HDx system from 16 Channels to 32 Channels.

Some sites that are upgrading to 32 Channel may need other components installed to make the system 32 Channel ready. These may have been shipped prior to this upgrade or shipped with this upgrade and installed at the same time. Refer to the following Catalog flow charts:

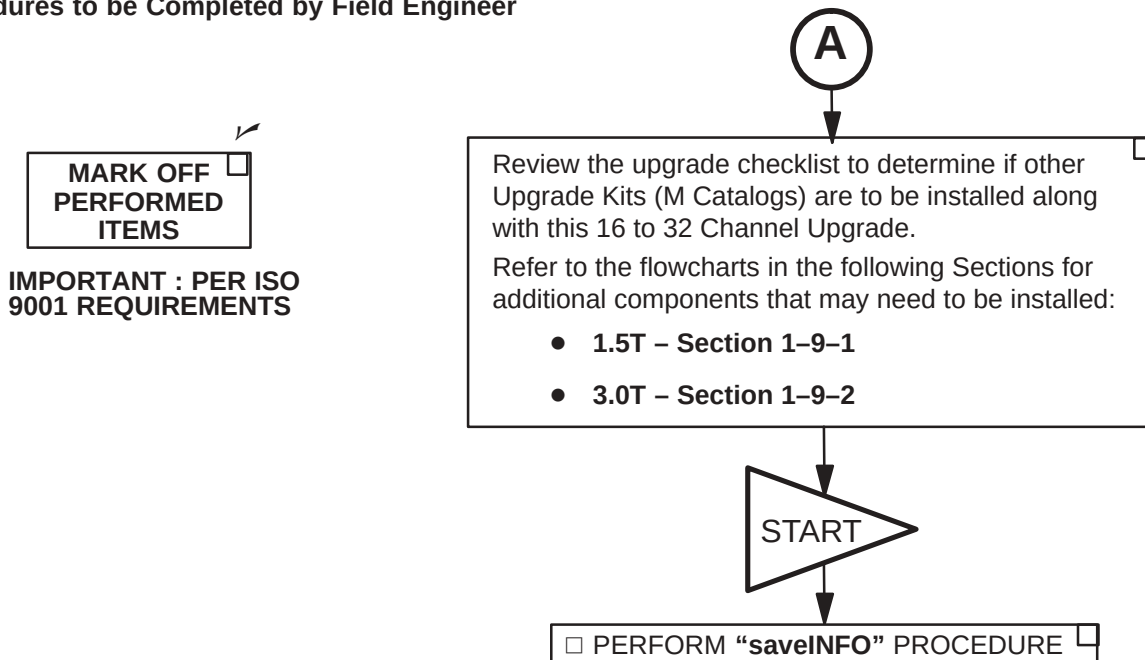
- 1.5T on page 11
- 3.0T on page 13

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

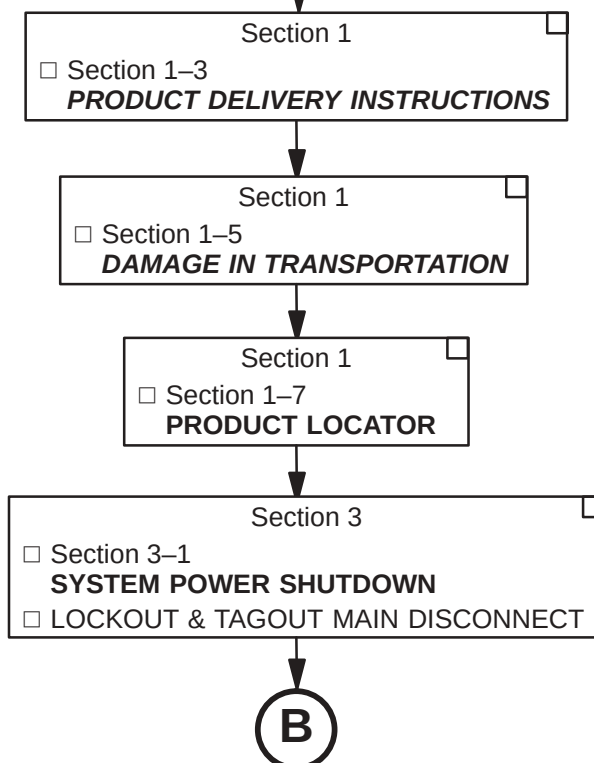
This upgrade flowchart has been developed assuming that all Signa HDx 32 Channel components have been delivered complete and together.

2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART

Procedures to be Completed by Field Engineer

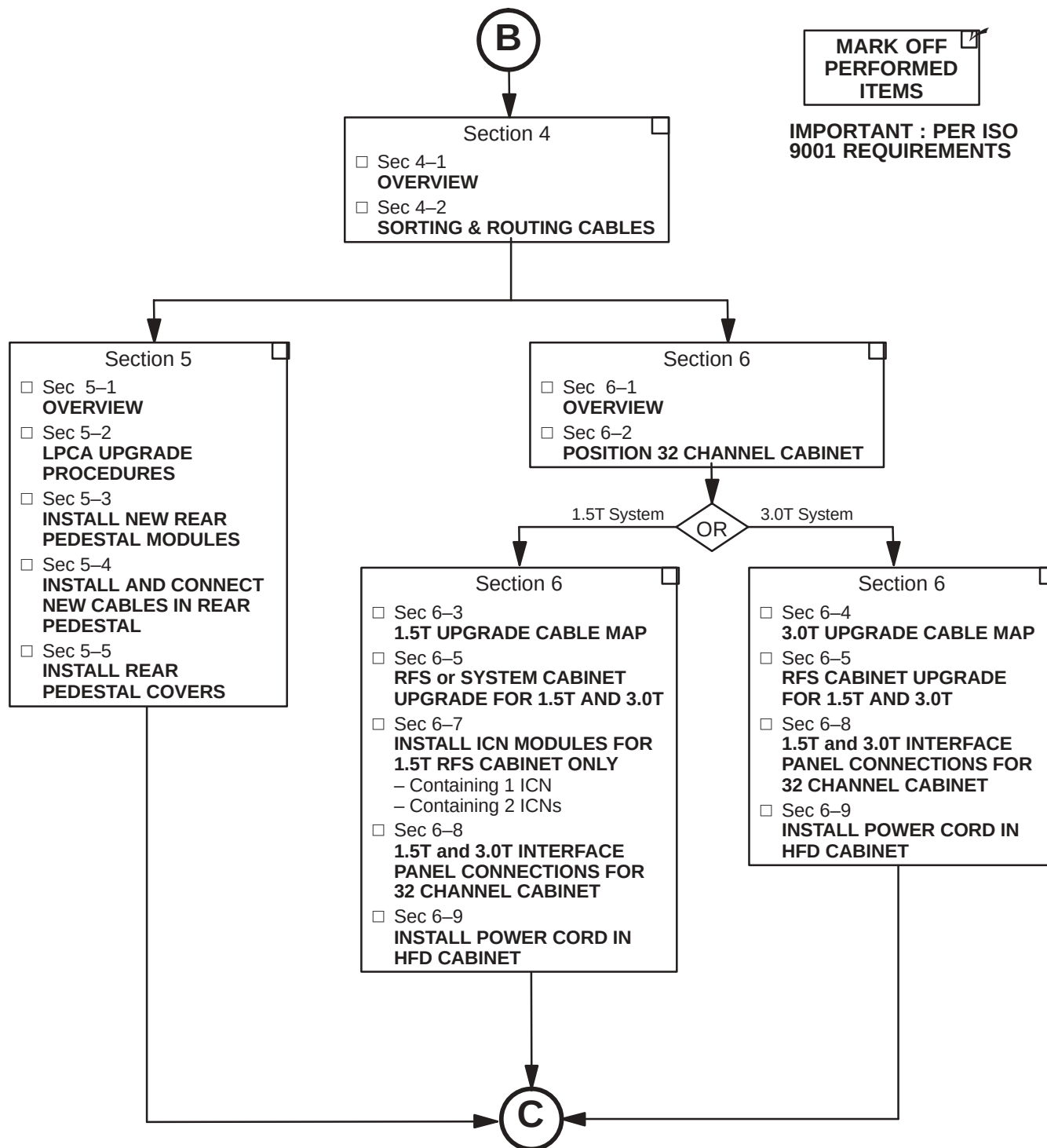


Procedures to be Completed by FE and/or Mechanical Installers



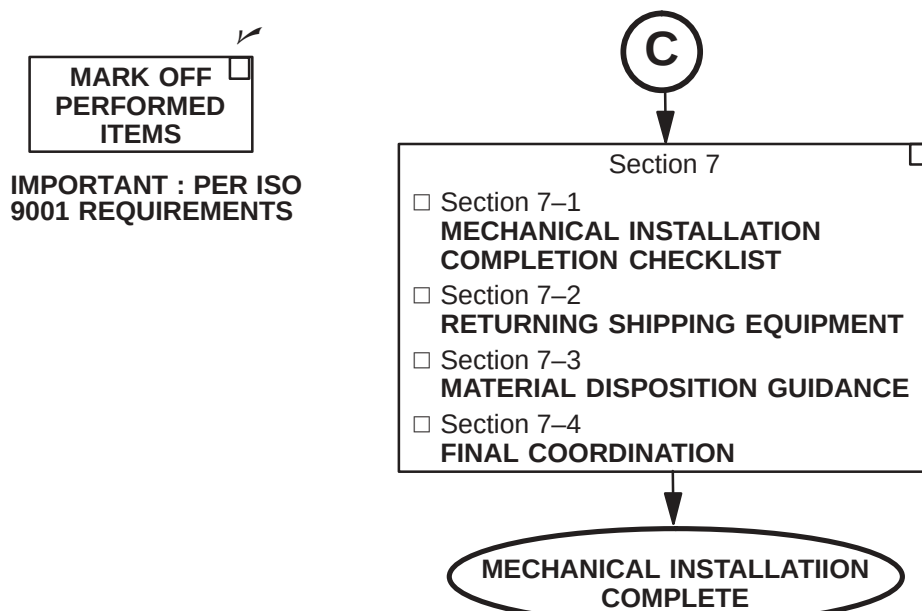
2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

De-Installation Procedures to be Completed by FE and/or Mechanical Installers

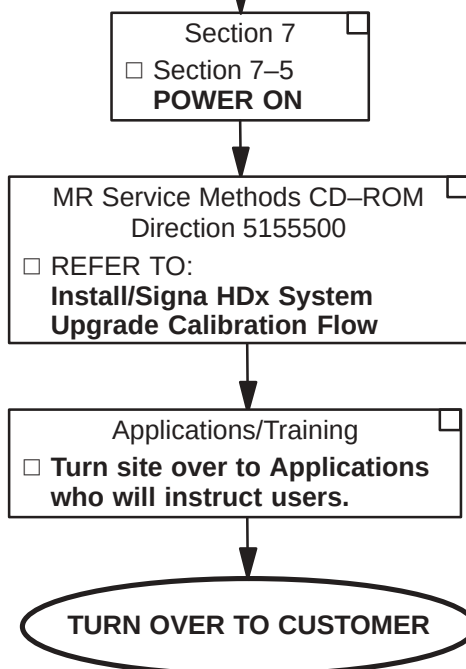


2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers



Procedures to be Completed by FE and Applications



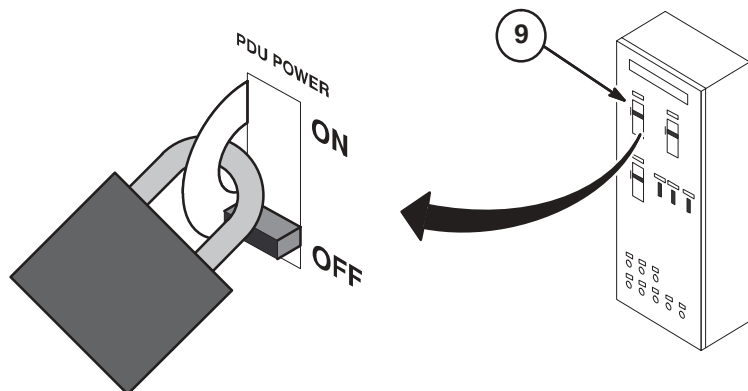
SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
 - IIS
2. Make sure that site customer has archived all patient information. Execute "Save Info".
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Perform normal software shutdown procedures.
6. Press the POWER OFF button on the PDU front control panel.
7. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
8. Locate and shut off main disconnect supplying power to the PDU.
9. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
10. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring



SECTION 4 – SYSTEM UPGRADE CABLES

4-1 OVERVIEW

This section provides information on routing and storing cables for upgrading a Signa HDx System to 32 Channels. Cable maps for the upgrade can be found at:

- Magnet Room for 1.5T or 3.0T – Section 5-4 Page 34
- Equipment Room for 1.5T – Section 6-3 Page 42
- Equipment Room for 3.0T – Section 6-4 Page 43

4-2 SORTING AND ROUTING CABLES



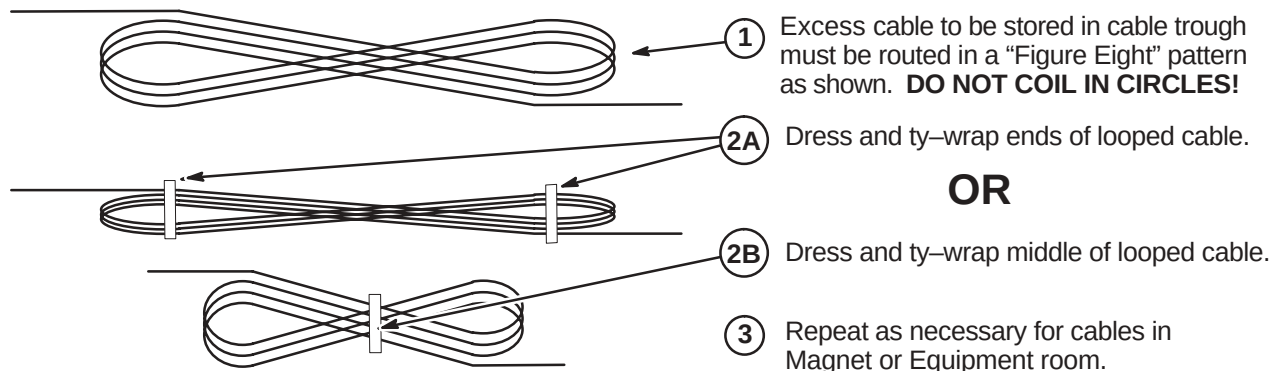
Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 4-1.



PROPER STORAGE OF EXCESS CABLES

ILLUSTRATION 4-1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

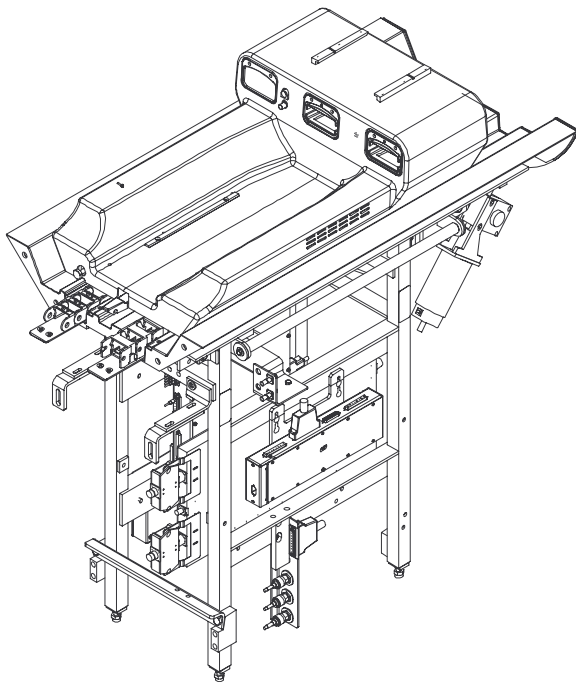
SECTION 5 – MAGNET ROOM UPGRADE

5-1 OVERVIEW

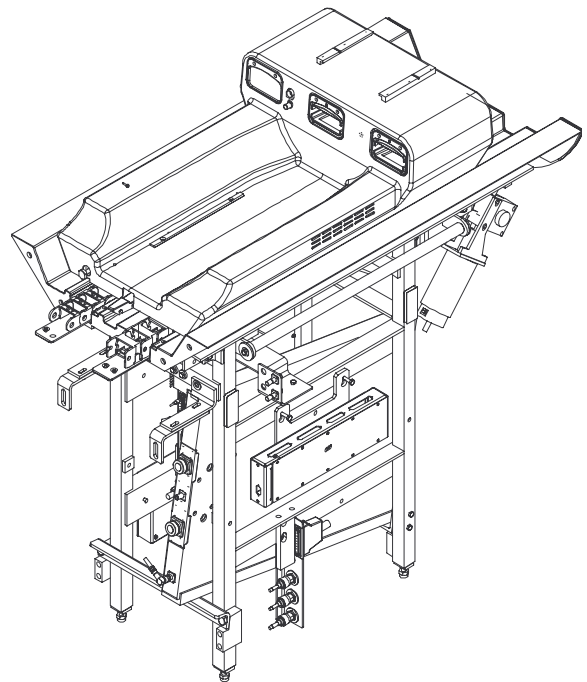
The following procedures are applicable to 1.5T and 3.0T Signa HDx configurations.

The LPCA will have three cables added to the larger cable track and they will connect to the new “C” connector to be installed on the LPCA. The LPCA upgrade procedures begin at Section 5-2.

The Rear Pedestal will provide a new 16 Channel Switch (#2), IRF Module (#2), and associated cables installed. The Rear Pedestal upgrade procedures begin at Section 5-3.



1.5T Rear Pedestal and LPCA prior to upgrade



3.0T Rear Pedestal and LPCA prior to upgrade

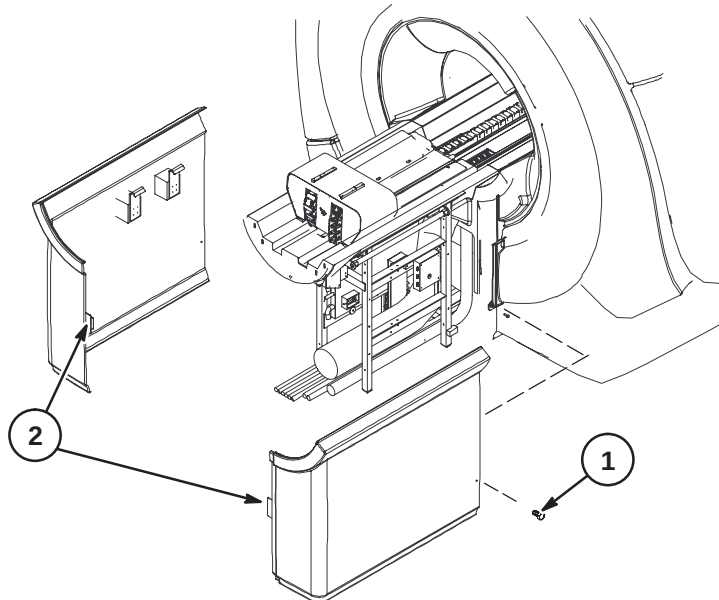
5-2 LPCA UPGRADE PROCEDURES

5-2-1 REMOVE REAR PEDESTAL COVERS AND DISCONNECT CABLES

The site being upgraded could have either the new HDx enclosure covers or the WideOpen enclosure covers. Both are shown in the following sub-sections.

5-2-1-1 HDx Cover Removal

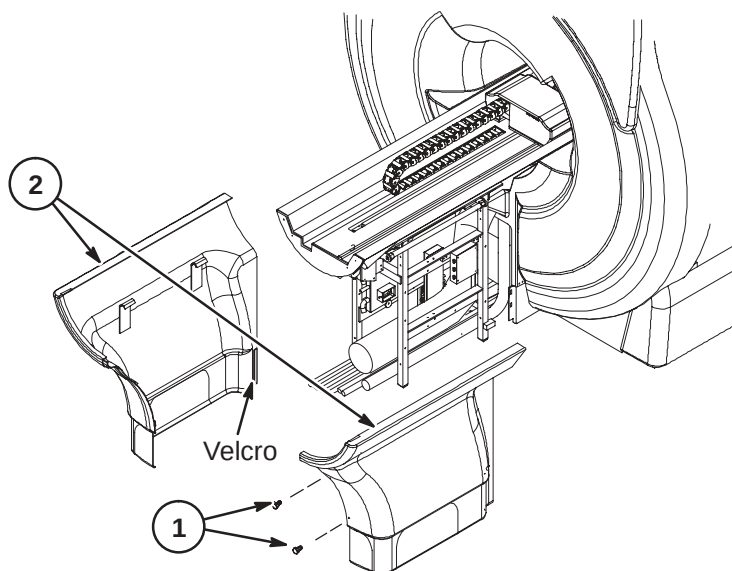
1. Unfasten Left and Right Pedestal Covers by removing screw on each cover.
2. Carefully remove Rear Pedestal covers and place in storage to prevent damage. If this is a new HDx system, they will be re-installed at completion of upgrade.

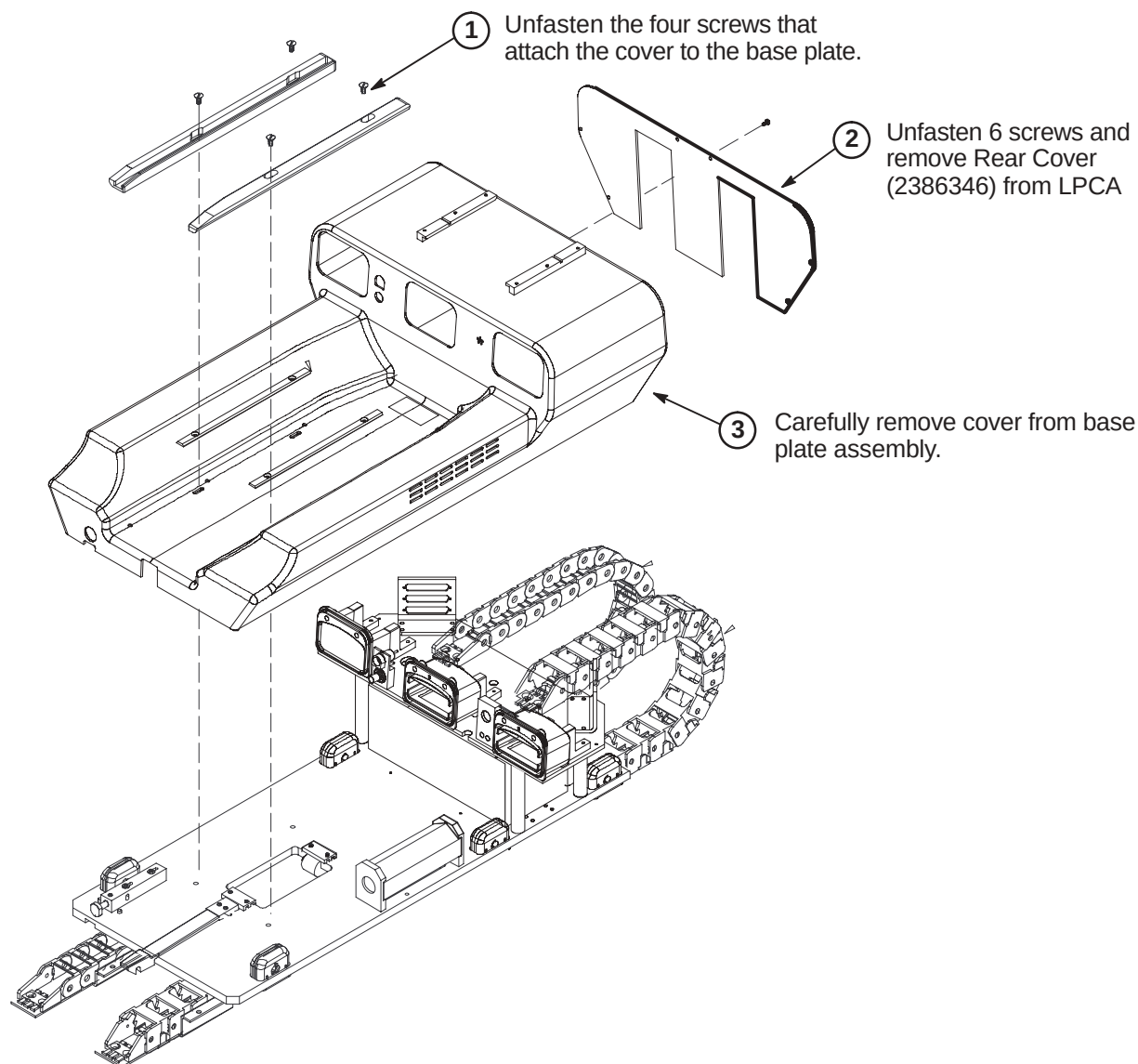


5-2-1-2 WideOpen Cover Removal

1. Unfasten screws holding covers together.
2. Lift covers off side bar and return for recycling.

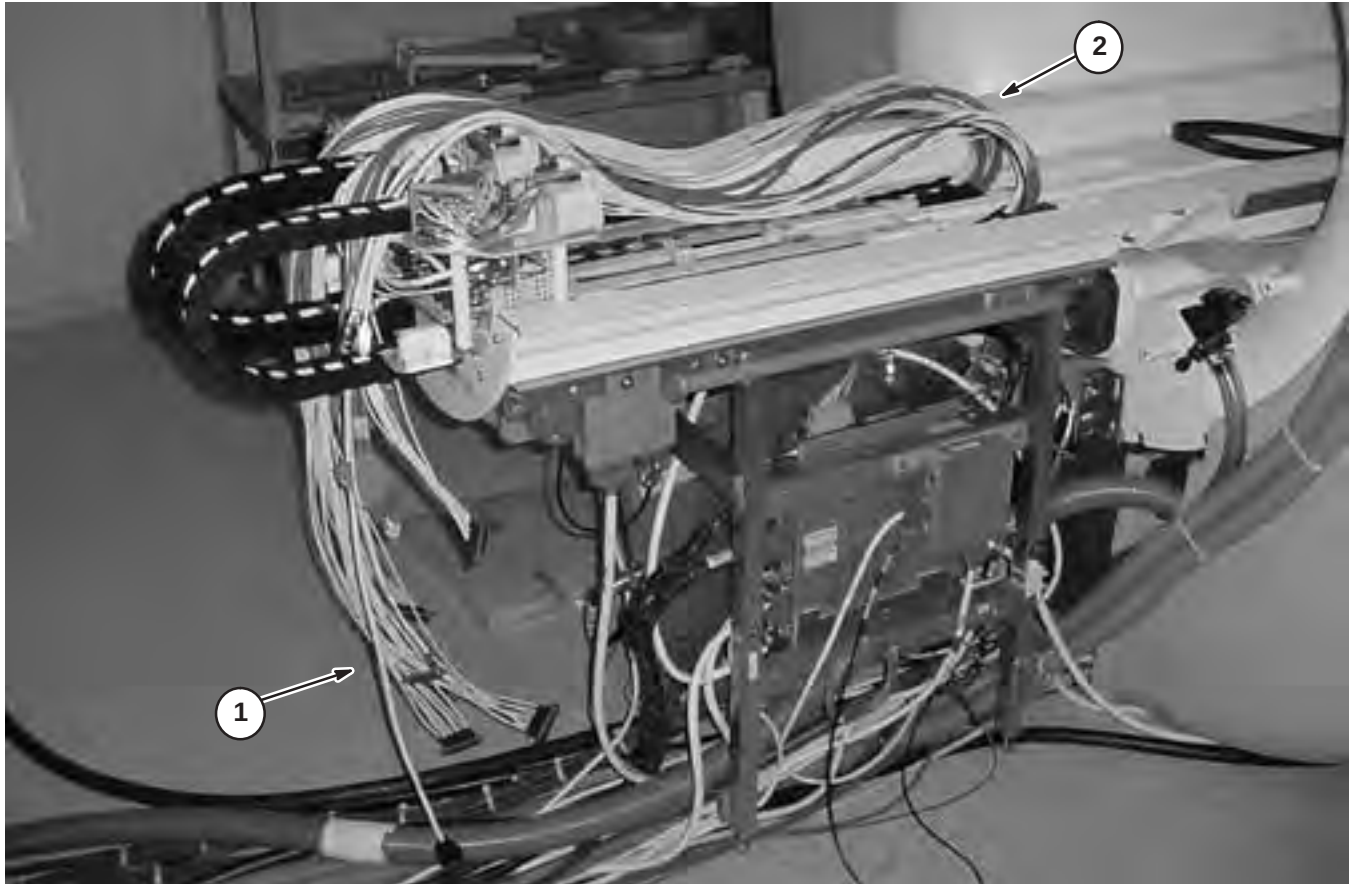
Note: Covers are also held in place with Velcro on lower part of covers as shown below.



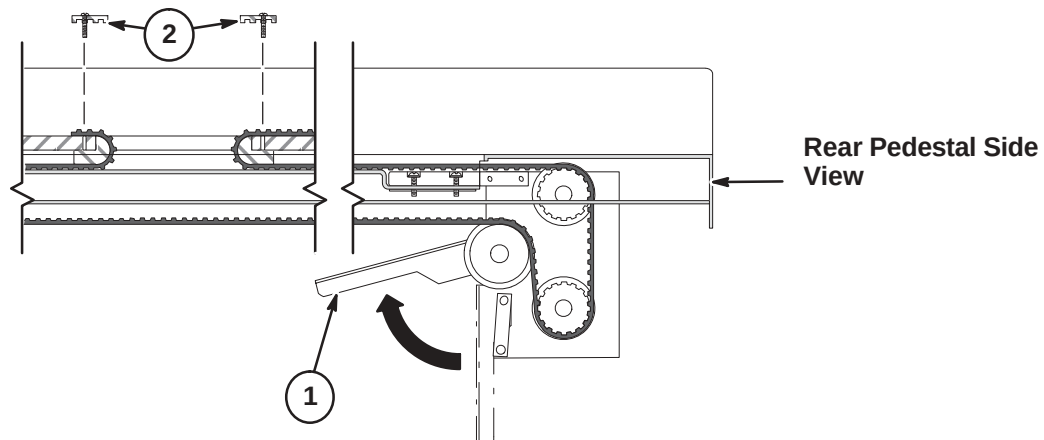
5-2-2 REMOVE LPCA COVER

5-2-3 Disconnect LPCA Cables

1. Disconnect all cables that are routed thru cable track and connected to modules in Rear Pedestal.
2. Carefully pull out disconnected cables thru holes in Bridge and place on LPCA as shown.

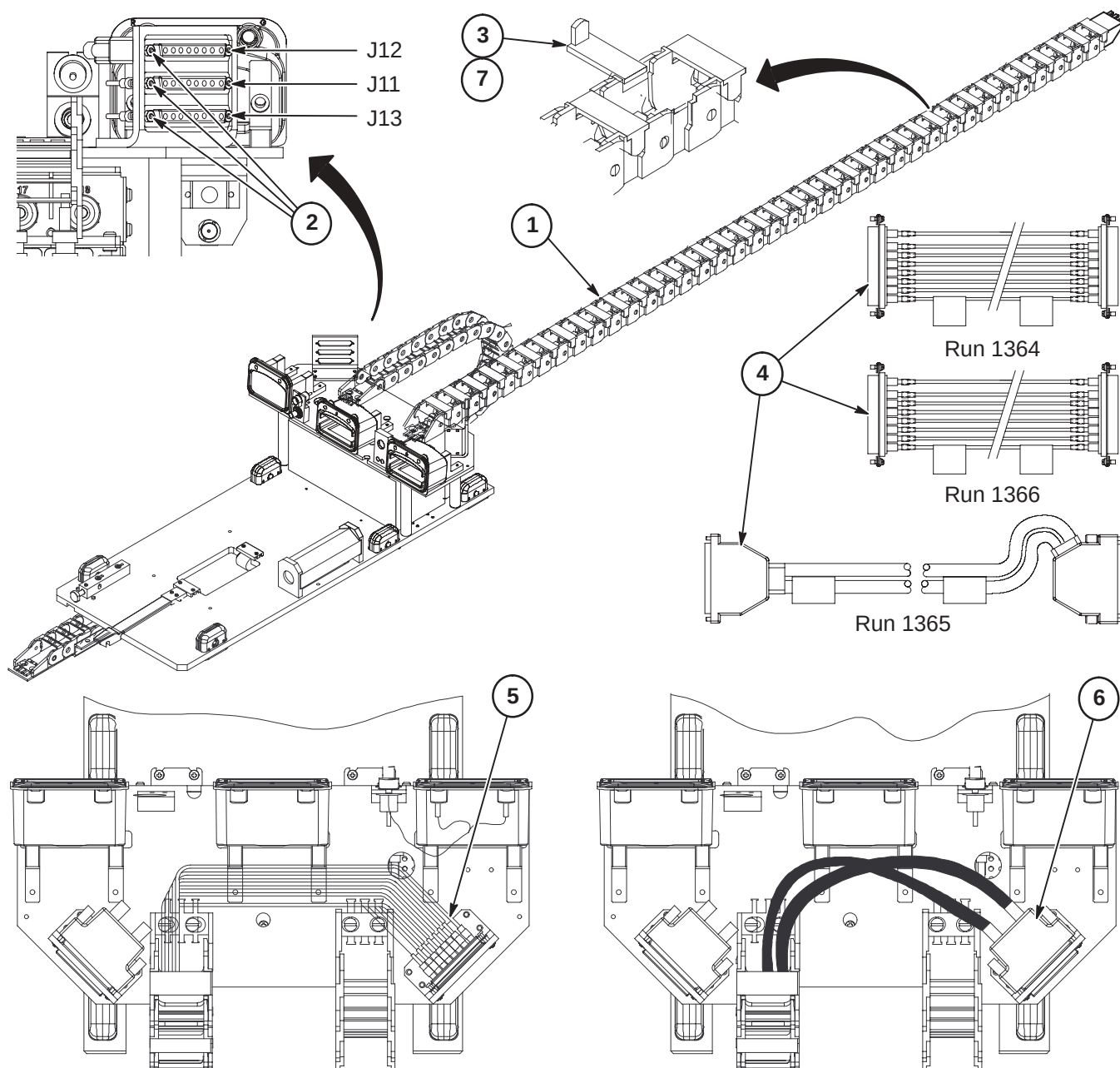
**5-2-4 Disconnect LPCA Drive Belt**

1. Release Tension Arm.
2. Unfasten screws, remove clamps, and disconnect belt.



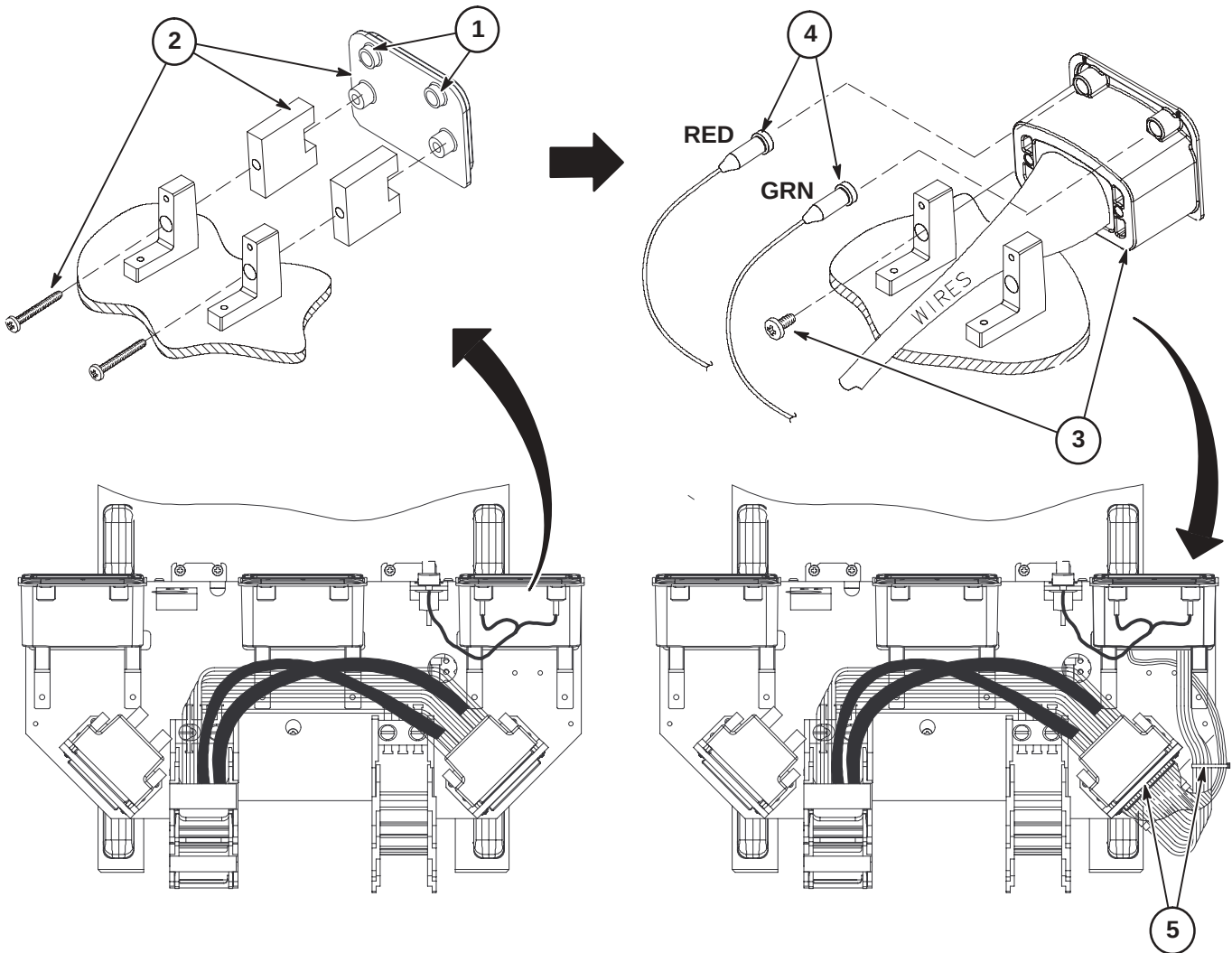
5-2-5 INSTALL RUNS 1364, 1365, AND 1366 IN CABLE TRACK

1. Move LPCA with attached Cable Tracks outside of magnet room and layout large cable track as shown.
2. On each connector, from the Jackscrew Kit (5117237), install two Hex Standoffs on the farside and two nuts on the nearside. DO NOT INSTALL WASHERS OR LOCKWASHERS. DISCARD THESE.
3. Flip open all of the top cross pieces of each section of the large Cable Track.
4. In the large Cable Track, place cable Runs 1364, 1365, and 1366. Refer to Section 5-4 for connecting the ends of the cables in the LPCA.
5. Connect Runs 1364 to J13 and 1366 to J11 bracket connector as shown.
6. Connect Run 1365 to J12 as shown.
7. Close top cross pieces after cables have been connected at LPCA end and dressed in Cable Track.



5-2-6 INSTALL "C" BEZEL AND CONNECT CABLES IN LPCA

1. Disconnect Red and Green LED wires from bezel cover.
2. Unfasten and discard the existing Bezel Cover, Spacer, and Screws.
3. Install C-Bezel and Cable harness (5111020-53) using two M5 x 20mm Brass Screws.
4. Connect Red and Green LED wires to new C-Bezel cover.
5. Connect C-Bezel Harness to J12, J11, and J13 as marked and tie-wrap as needed.



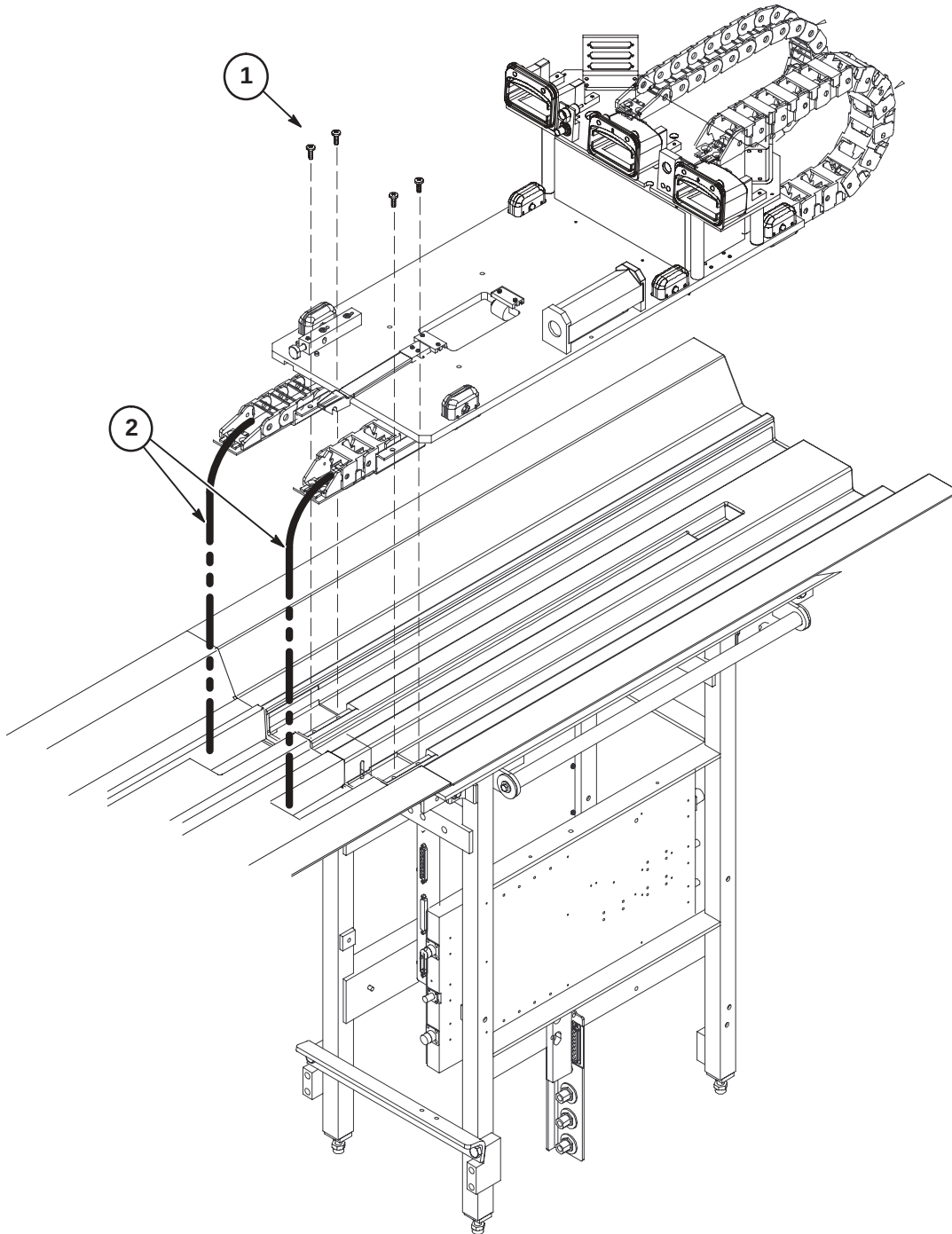
5-2-7 COMPLETED LPCA UPGRADE

The following picture shows the LPCA with new cables after the upgrade is complete.



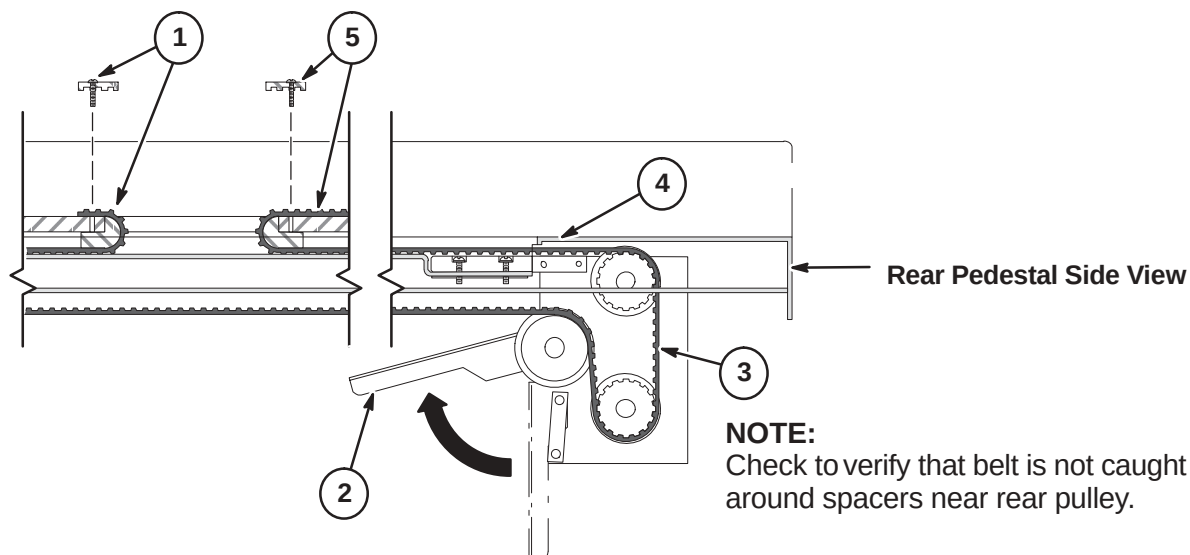
5-2-8 RE-INSTALL LPCA AND CABLE TRACK

1. Attach the Cable Track(s) to the Rear Split Bridge using four (or two for one cable track) M6 x 20mm Pan Head Nylon Screws (2378292).
2. Feed the cables from the cable track(s) thru the openings in the Front Split Bridge.

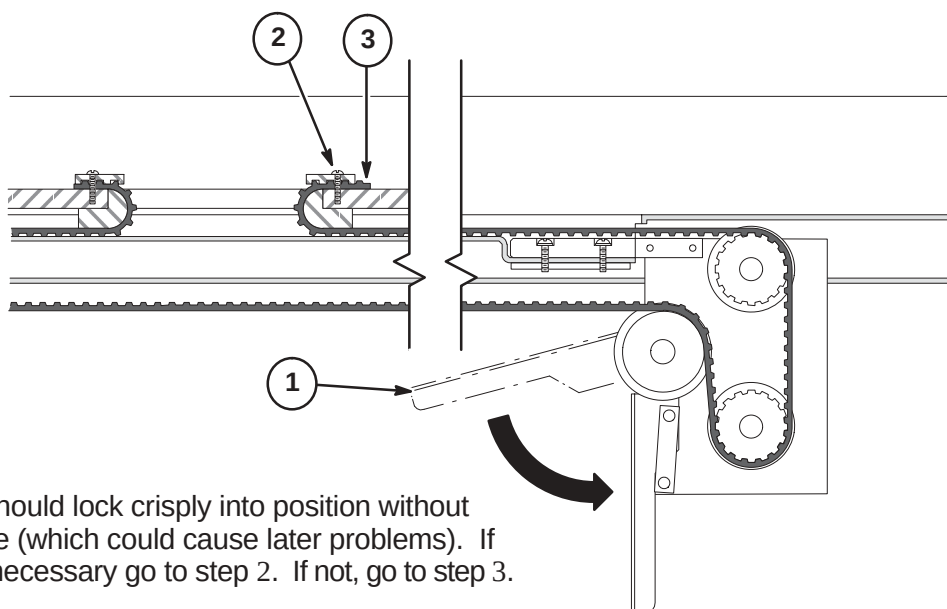


5-2-9 ROUTE AND ATTACH DRIVE BELT

1. Route one end of Drive Belt delivered with Front Split Bridge thru opening in LPCA Base Plate and attach with clamp.
2. Make sure Tension Arm is in release position.
3. Feed back end of Belt underneath Bridge, around Tension Arm Pulley and rear pulleys into Bridge.
4. Route Belt under and through opening to top of Carriage.
5. Position Belt under Clamp Plate and attach.

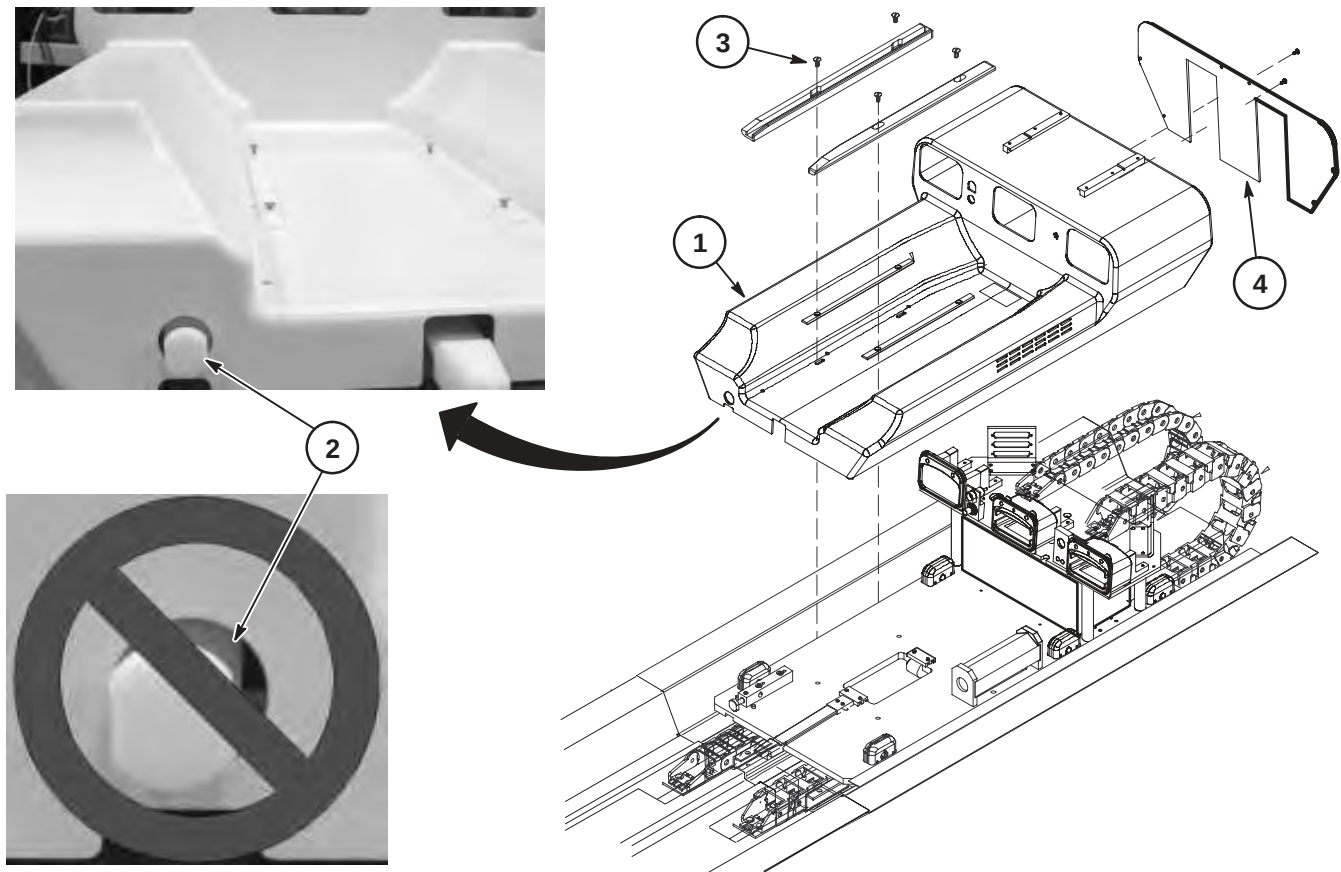
**5-2-10 FINAL ADJUSTMENT OF DRIVE BELT**

1. Check Belt tension by closing Tension Arm. (See Note)
2. **(IF NECESSARY):** Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.
3. Tighten Clamps and trim excess Belt length.



5-2-11 INSTALL LPCA COVER

1. Carefully place cover on base plate assembly.
2. Center the front knob within the hole on the x-axis. The y-axis does not need to be centered.
3. Once centered, secure the cover to the base plate using four screws previously removed.
4. Attach Rear Cover using 6 screws previously removed.

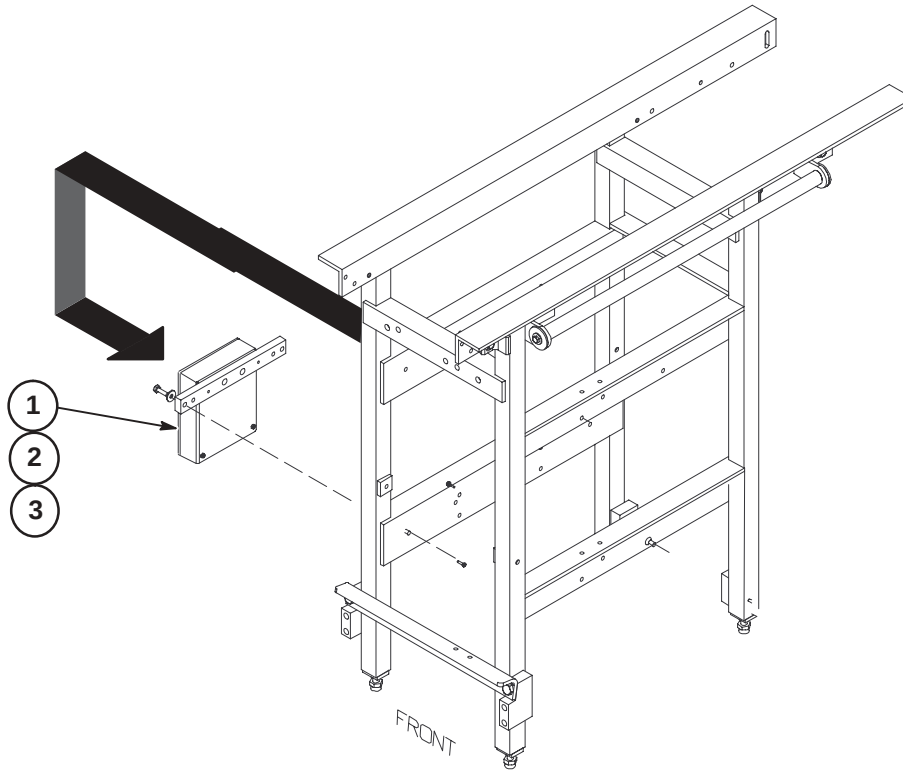


5-3 INSTALL NEW REAR PEDESTAL MODULES

The same procedures are required for upgrading the 1.5T or 3.0T Rear Pedestals. Therefore, only the 1.5T Rear Pedestal will be shown in the following upgrade steps.

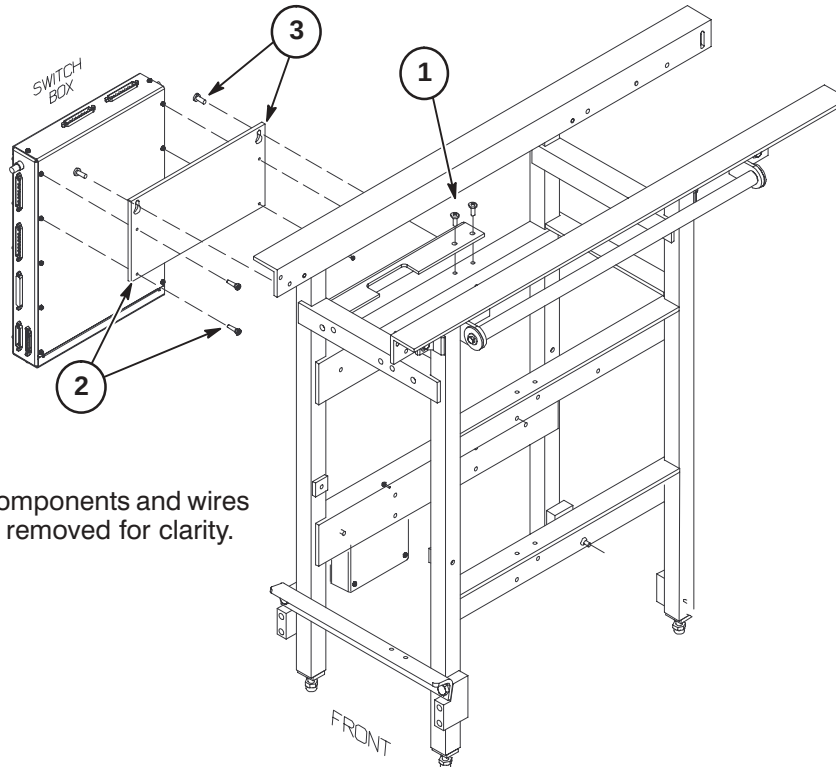
5-3-1 RELOCATE PATIENT COMMUNICATION BOX

1. Disconnect cables from Patient Communication Box.
2. Relocate Patient Communication Box from upper to lower horizontal cross brace.
3. Reconnect cables to Patient Communication Box.



5-3-2 INSTALL SECOND 16 CHANNEL SWITCH

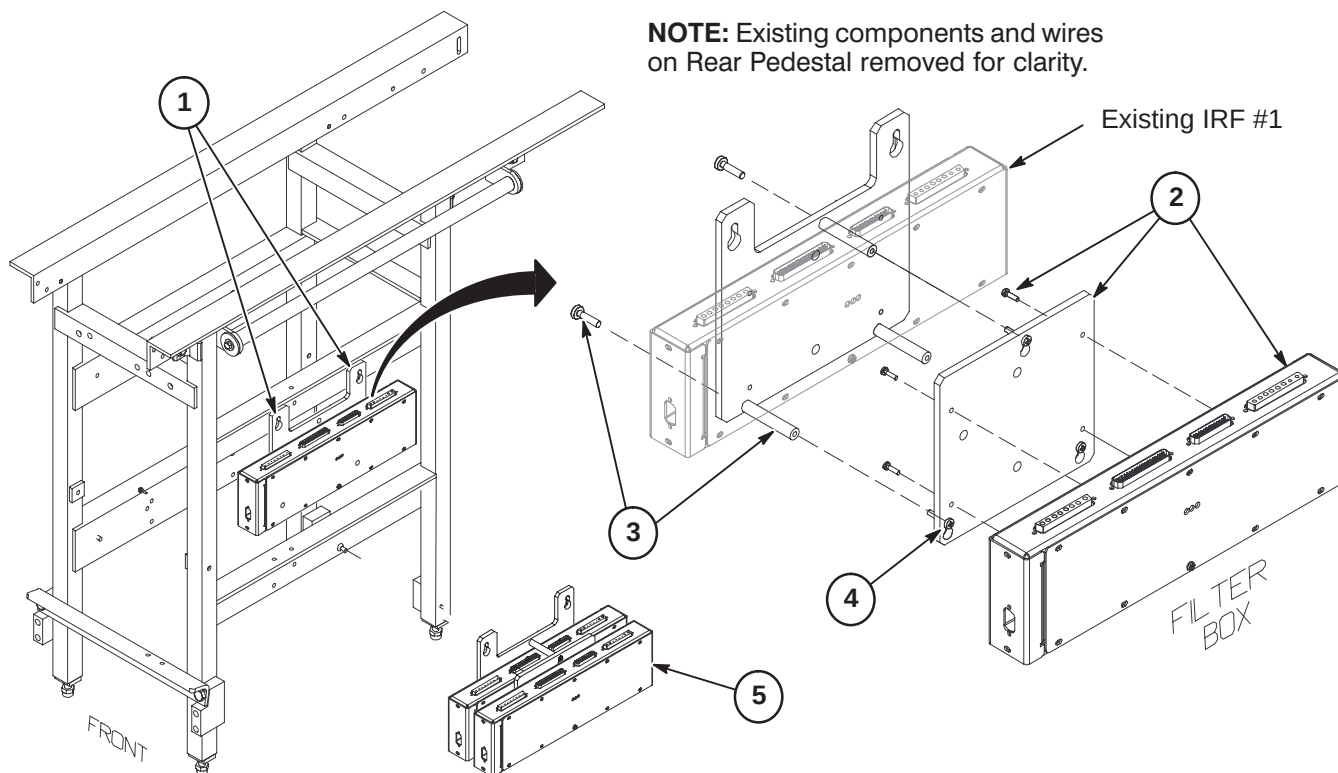
1. Install the 16 Channel Switch Mounting Angle (5139295) to top cross brace using two M6 x 16mm (2109873-25) Pan Head screws.
2. Attach 16 Channel Switch Outer Mounting Plate (5139296) to 16 Channel Switch #2 (5109645-3 for 1.5T, 5109645-4 for 3.0T) using four M3 x 12mm Phillips Cross screws (46-312342P24).
3. Attach Switch and Plate assembly to Angle using two M6 x 16mm (2109873-25) Pan Head screws.



NOTE: Existing components and wires on Rear Pedestal removed for clarity.

5-3-3 INSTALL IMAGE REJECT FILTER #2

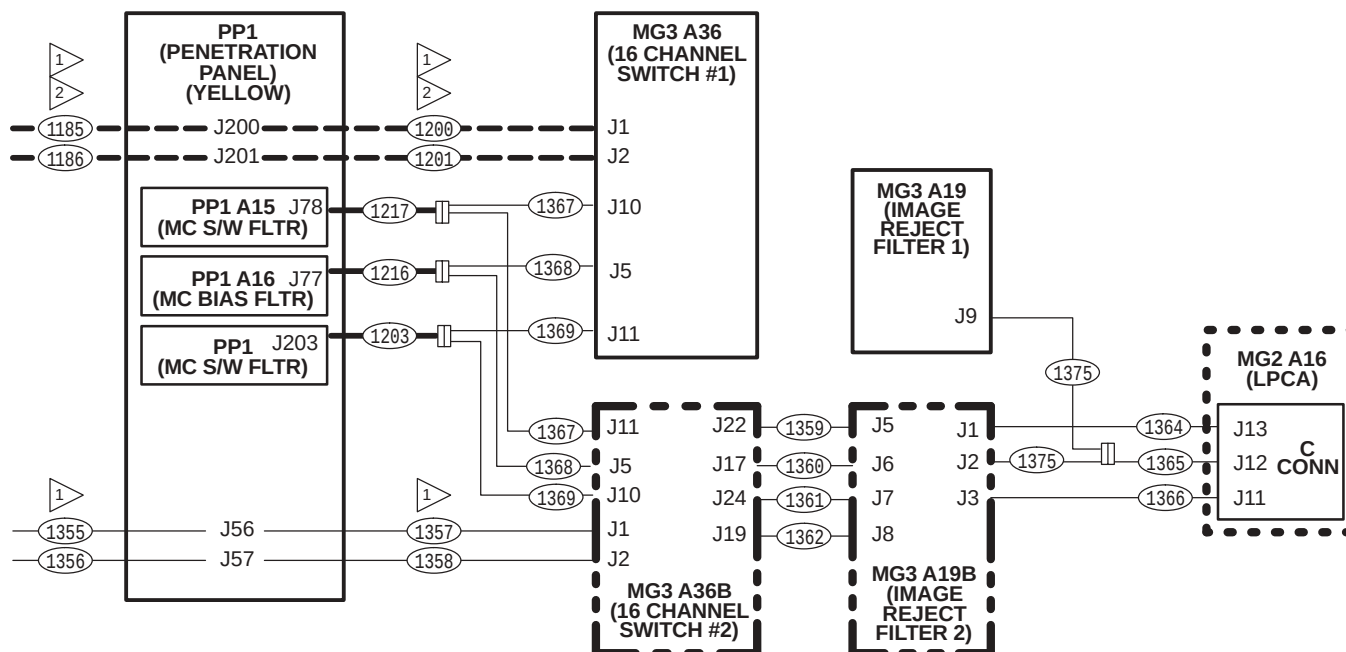
1. If necessary, for easier assembly, unfasten existing IRF mounting plate with IRF #1 from Rear Pedestal frame.
2. Attach IRF Outer Mounting plate (5139298) to Image Reject Filter #2 (5137348 for 1.5T, 5137348-2 for 3.0T) using four M3 x 12mm Phillips Cross screws (46-312342P24).
3. Attach three Standoffs (5139297) to IRF #1 Mounting plate using three M4 x 20mm Brass Slotted Head screws (5120369-65).
4. Attach IRF #2/Mounting Plate assembly to Standoffs using three M4 x 20mm Pan Head screws (2109873-17).
5. View of final assembly.



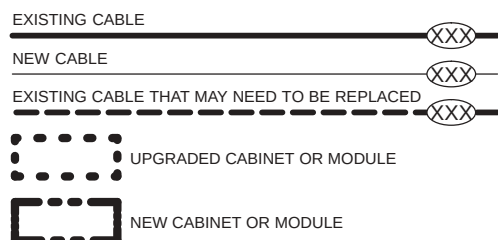
5-4 INSTALL AND CONNECT NEW CABLES IN REAR PEDESTAL

The cable map below indicates the new cables to be routed and connected in the Magnet Room. It also shows three existing cables that will be disconnected at the end connected to the 16 Channel Switch #1 and reconnected to three new Y-cables.

Refer to the following procedures for installing these cables.



LEGEND:



NOTES:

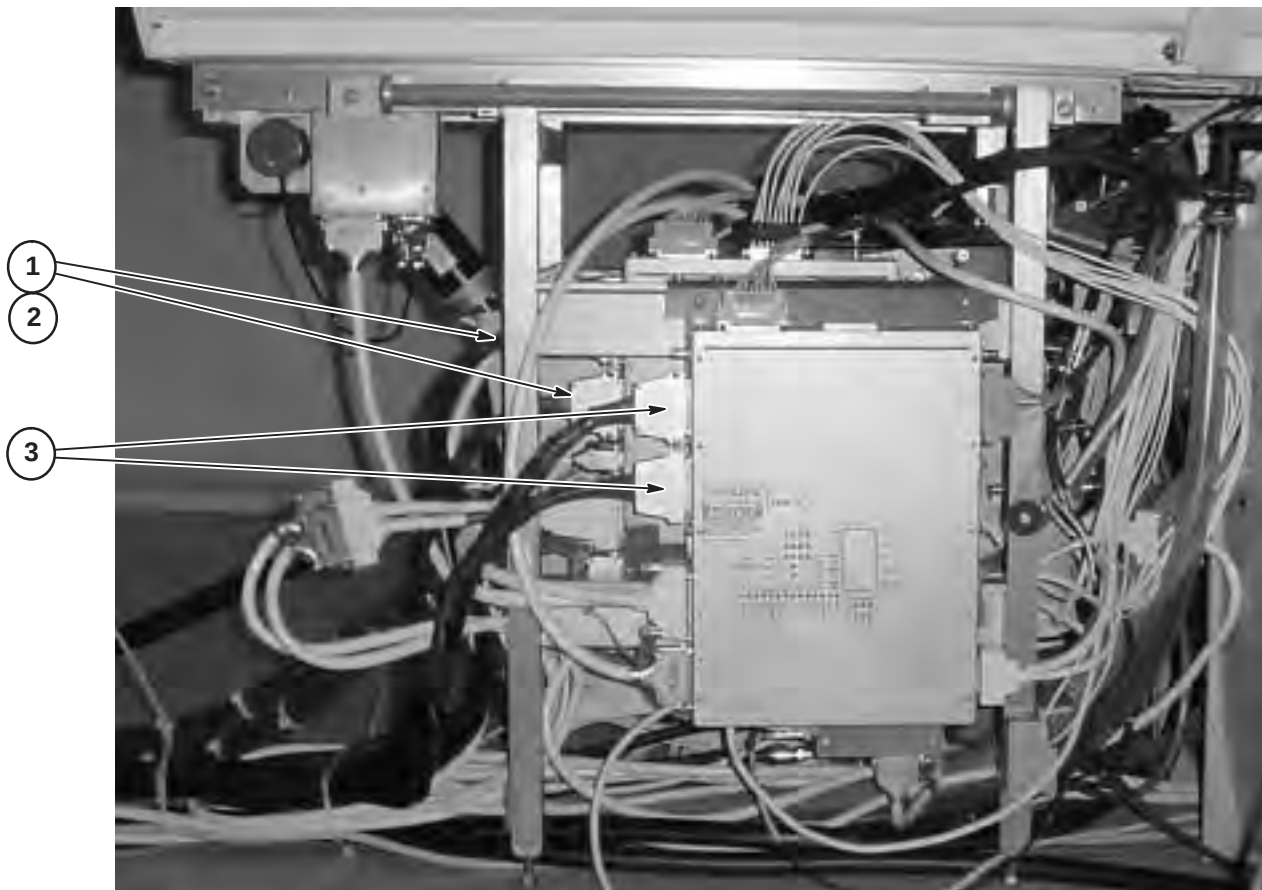
- 1 Runs 1185 and 1186 MUST be the same length as Runs 1355 and 1356. Runs 1200 and 1201 MUST be the same length as Runs 1357 and 1358.
- 2 Refer to Section 5-4-1 for instructions to determine if Runs 1200 and 1201 are required to be replaced.

5-4-1 INSTALL RUNS 1357 AND 1358 AT 16 CHANNEL SWITCH #2**Note**

For system to run at optimum performance, existing Runs 1200 and 1201 **MUST** be the same length as new Runs 1357 and 1358 in the magnet room. Also, Runs 1185 and 1186 **MUST** be the same length as new Runs 1355 and 1356 in equipment room.

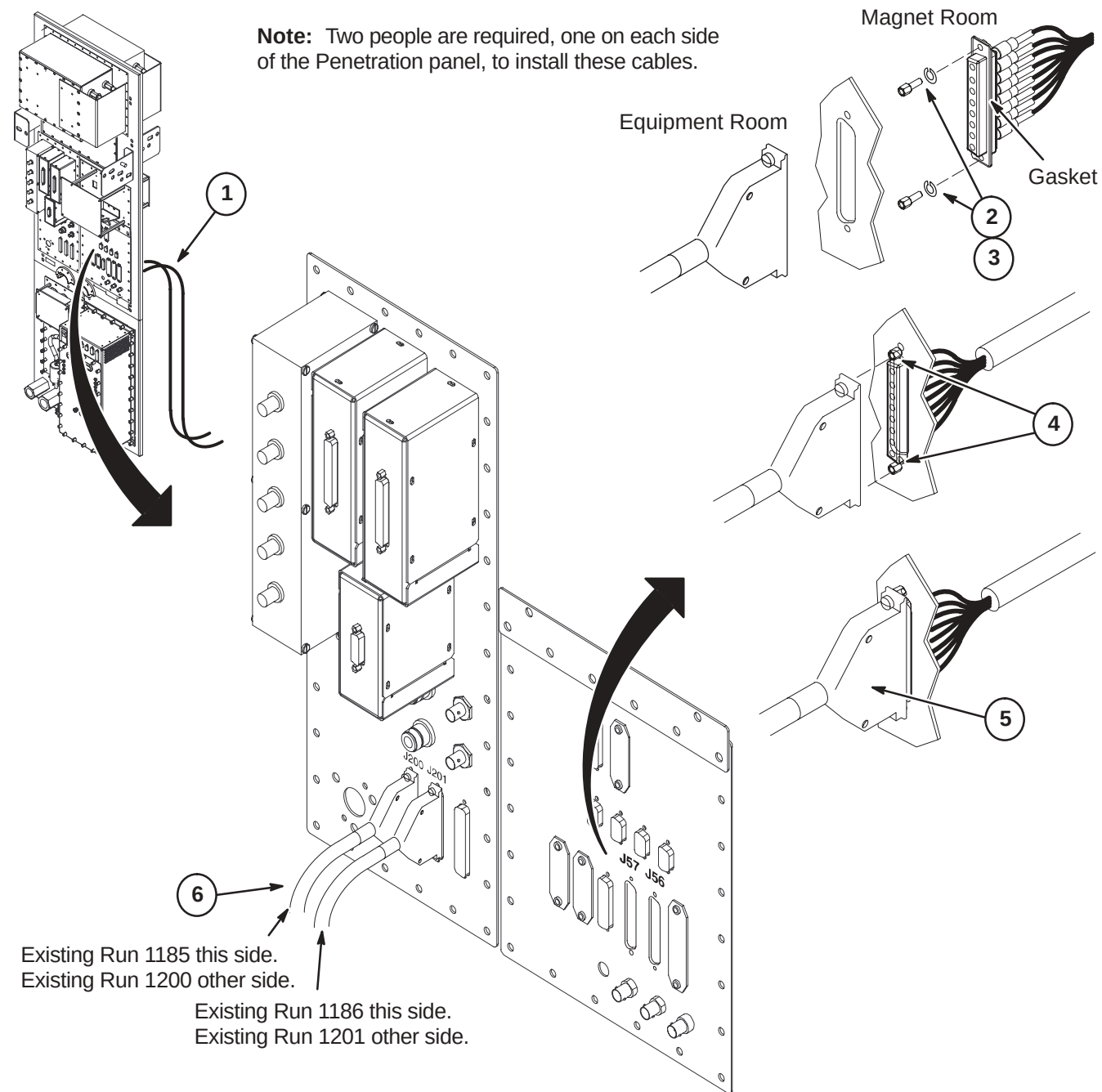
The 32 Channel Upgrade kit provides long (24400mm/80 ft.) and short (10515mm/34.5 ft.) lengths of each of the following cable Runs:

- Runs 1185 and 1186 (cables are already installed on site in equipment room)
 - Runs 1200 and 1201 (cables are already installed on site in magnet room)
 - Runs 1355 and 1356 (new cables to be installed in equipment room)
 - Runs 1357 and 1358 (new cables to be installed in magnet room)
1. If possible, determine the length of existing Runs 1200 and 1201.
 - A. If length is same as either of the long or short lengths of new Runs 1357 and 1358 that are being installed, they may be retained.
 - B. If length can not be determined, de-install cables and return for recycling.
 2. If required, install new Runs 1200 and 1201 that are the same length as the new runs 1357 and 1358.
 3. Connect new Runs 1357 and 1358 to J1 and J2 on new 16 Channel Switch #2 and route to Penetration Panel.



5-4-2 CONNECT RUNS 1357 AND 1358 TO PENETRATION PANEL

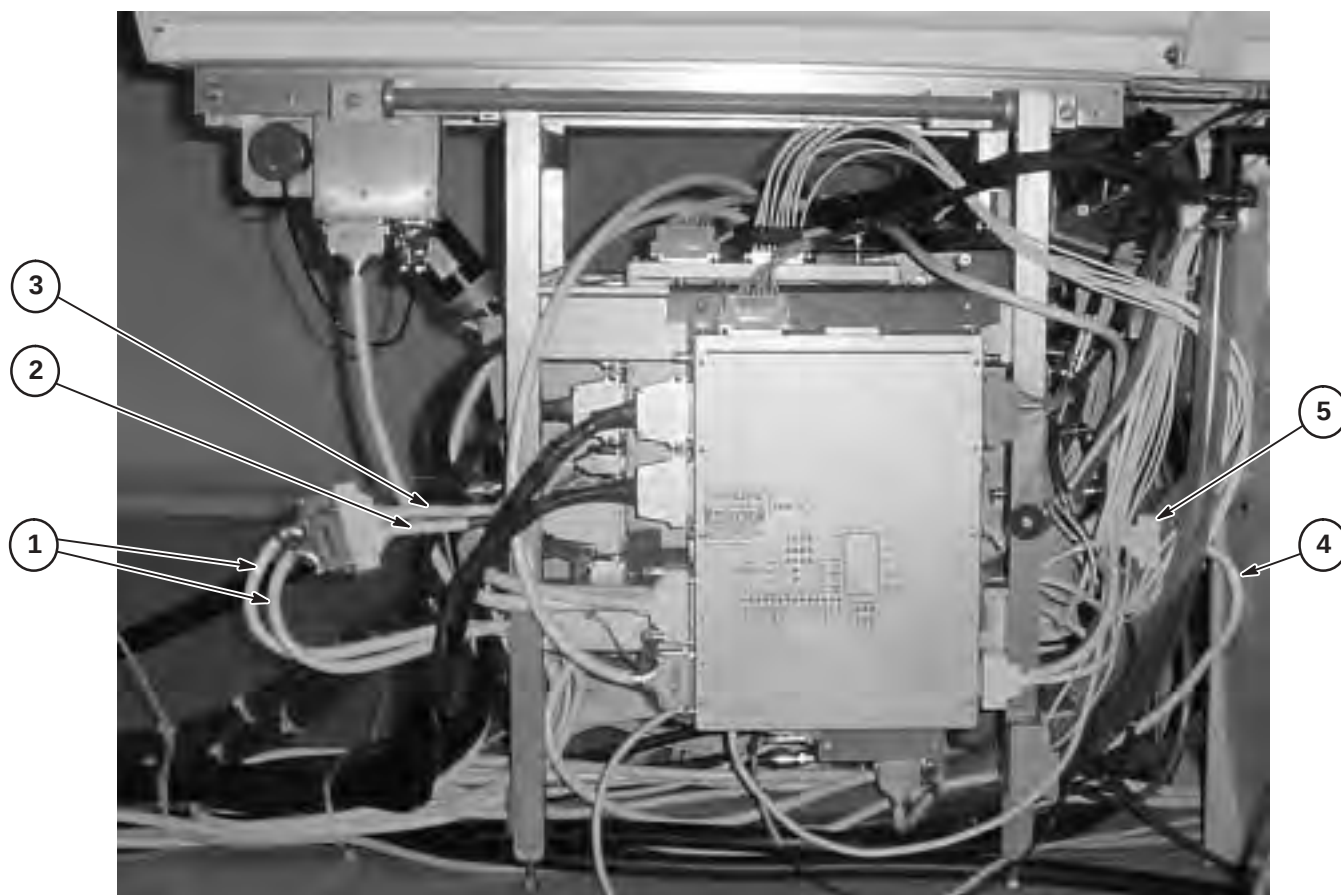
1. Route Runs 1357 and 1358 to Magnet Room side of Penetration Panel.
2. At J56 and J57: Remove jackscrews and lockwashers from housing plugs of Runs 1357 and 1358.
NOTE: Discard lockwashers. Do not use lockwashers to attach cables to RF panel.
3. Discard the lockwashers.
4. Using jackscrews, mount plugs of Runs 1357 and 1358 to RF panel as marked.
5. Connect Run 1357 to Run 1355 at J56 and Run 1358 to Run 1356 at J57.
6. If required, repeat the above steps to install new cables at J200 and J201. Refer to Section 5-4-1.



5-4-3 INSTALL Y-CABLES TO 16 CHANNEL SWITCHES

Note: The cable ends marked "16 Channel Switch" on each of the following Y-cables are interchangeable and can be connected to either 16 Channel Switch module. Refer to cable map in Section 5-4.

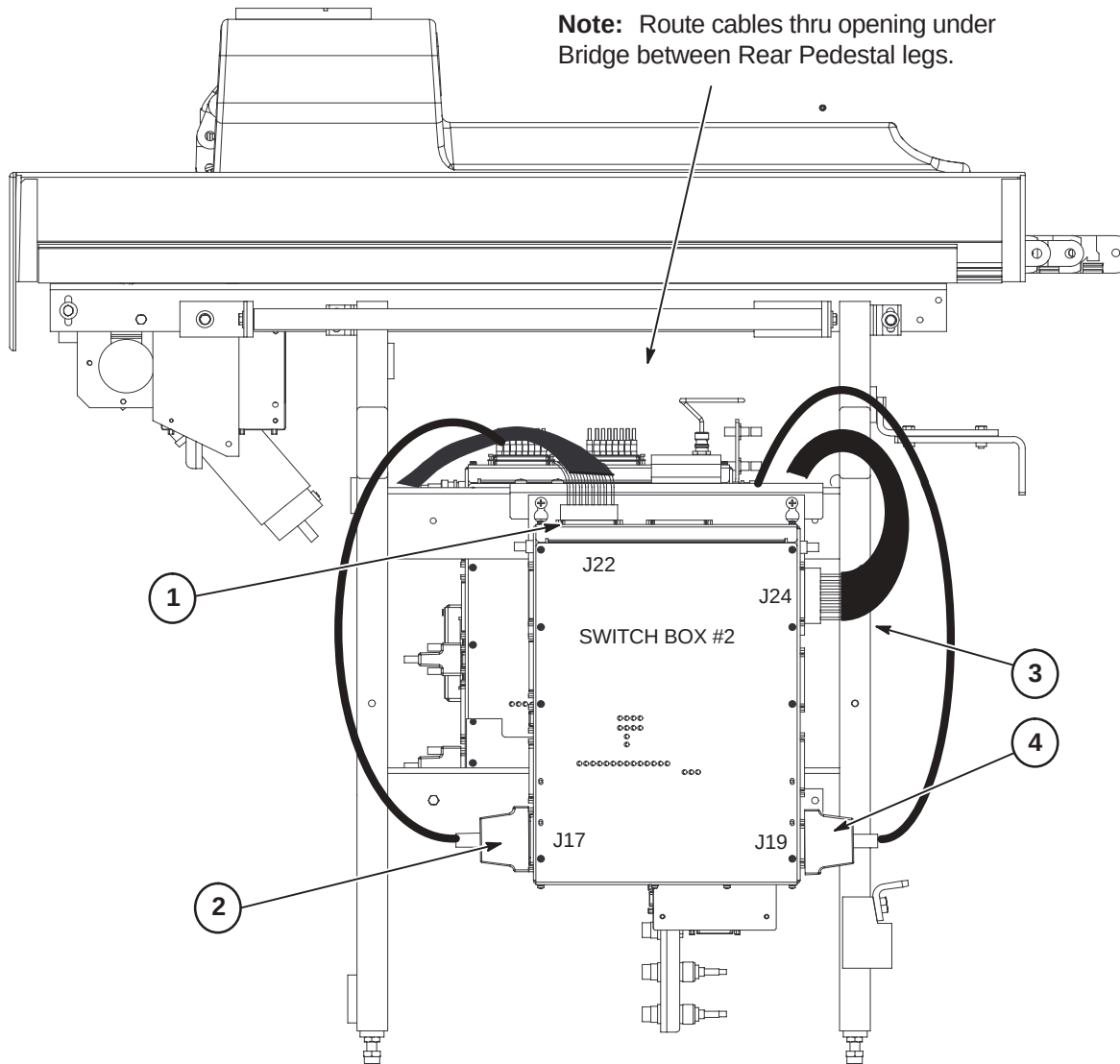
1. Disconnect Runs 1203 from J11 and run 1217 from J10 on existing 16 Channel Switch #1.
2. As marked on cable ends, install Y-Cable Run 1369 (5141870-2) by connecting to J11 on 16 Channel Switch #1, J10 on 16 Channel Switch #2, and Run 1203.
3. As marked on cable ends, install Y-Cable Run 1367 (5141870) by connecting to J10 on 16 Channel Switch #1, J11 on 16 Channel Switch #2, and Run 1217.
4. Disconnect Runs 1216 from J5 on existing 16 Channel Switch #1.
5. As marked on cable ends, install Y-Cable Run 1368 (5141871) by connecting to J5 on 16 Channel Switch #1, J5 on 16 Channel Switch #2, and Run 1216.



5-4-4 CONNECT REMAINING CABLES TO 16 CHANNEL SWITCH #2

1. Connect Run 1359 (5141244-3) to J22 and route to Image Reject Filter #2
2. Connect Run 1360 (5141362-2) to J17 and route to Image Reject Filter #2
3. Connect Run 1361 (5141244-4) to J24 and route to Image Reject Filter #2
4. Connect Run 1362 (5141363-2) to J19 and route to Image Reject Filter #2

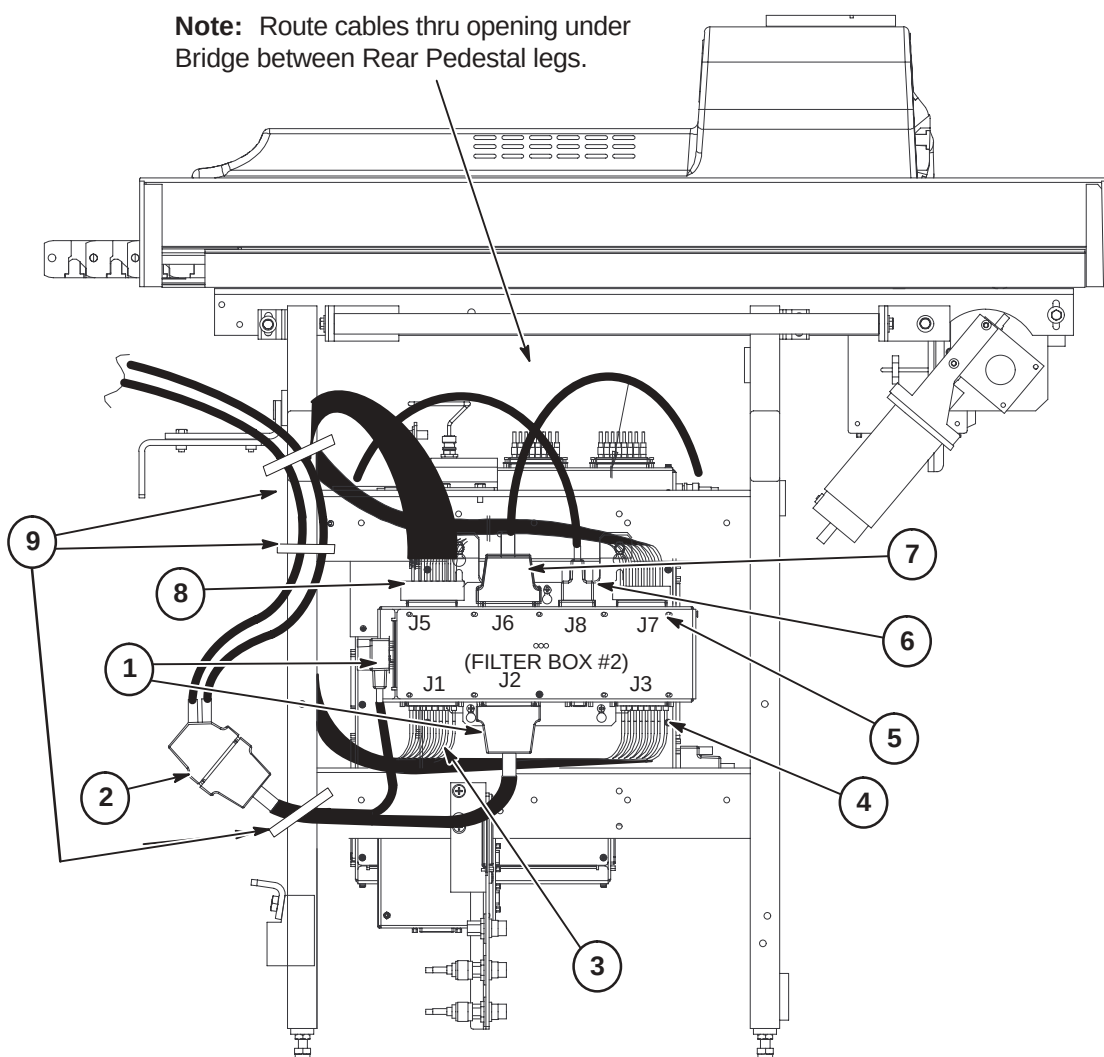
Note: Route cables thru opening under Bridge between Rear Pedestal legs.



5-4-5 CONNECT CABLES TO IMAGE REJECT FILTER #2

1. As marked on cable ends, install Y-Cable Run 1374 (5143204) by connecting to J9 on Image Reject Filter #1, J2 on Image Reject Filter #2, and Run 1365.
2. Route Run 1365 to Y-cable and connect as marked.
3. Route and connect Run 1364 from LPCA to J1 on IRF #2.
4. Route and connect Run 1366 from LPCA to J3 on IRF #2.
5. Route and connect Run 1361 from 16 Channel Switch #2 to J7 on IRF #2.
6. Route and connect Run 1362 from 16 Channel Switch #2 to J8 on IRF #2.
7. Route and connect Run 1360 from 16 Channel Switch #2 to J6 on IRF #2.
8. Route and connect Run 1359 from 16 Channel Switch #2 to J5 on IRF #2.
9. Install tie-wraps (46-208758P5) as shown to secure cables.

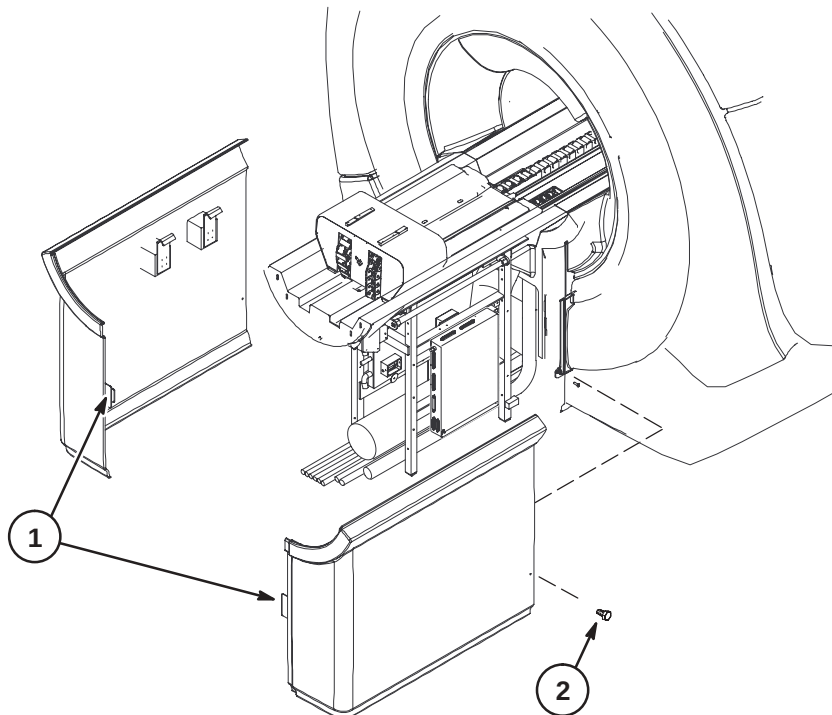
Note: Route cables thru opening under Bridge between Rear Pedestal legs.



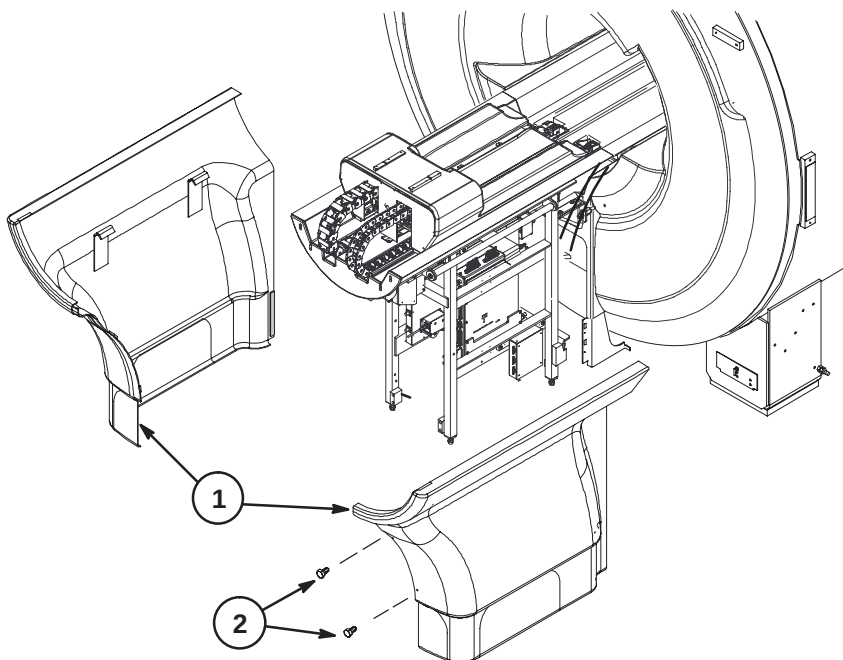
5-5 INSTALL REAR PEDESTAL COVERS

Steps 1 and 2 are for HDx. Steps 3 and 4 are for Wideopen.

1. Move Rear Pedestal covers into place and snap together.
2. Secure Left and Right Pedestal Covers by fastening screw on each cover. (for new HDx enclosure covers)



3. Hang new left and right Rear Pedestal Covers (5180369 & 5178111) on horizontal support bars.
4. Secure left and right covers by fastening two M6 x 10mm Pan Head screws as shown below.



SECTION 6 – EQUIPMENT ROOM UPGRADE

6-1 OVERVIEW

The following procedures are applicable to 1.5T and 3T Signa HDx or HDxt configurations.

There are multiple existing ICN configurations. All will upgrade to a one ICN configuration, part number **5337894-4** for the 32 channel system.

NOTE: There are two different numbers for the latest ICN:

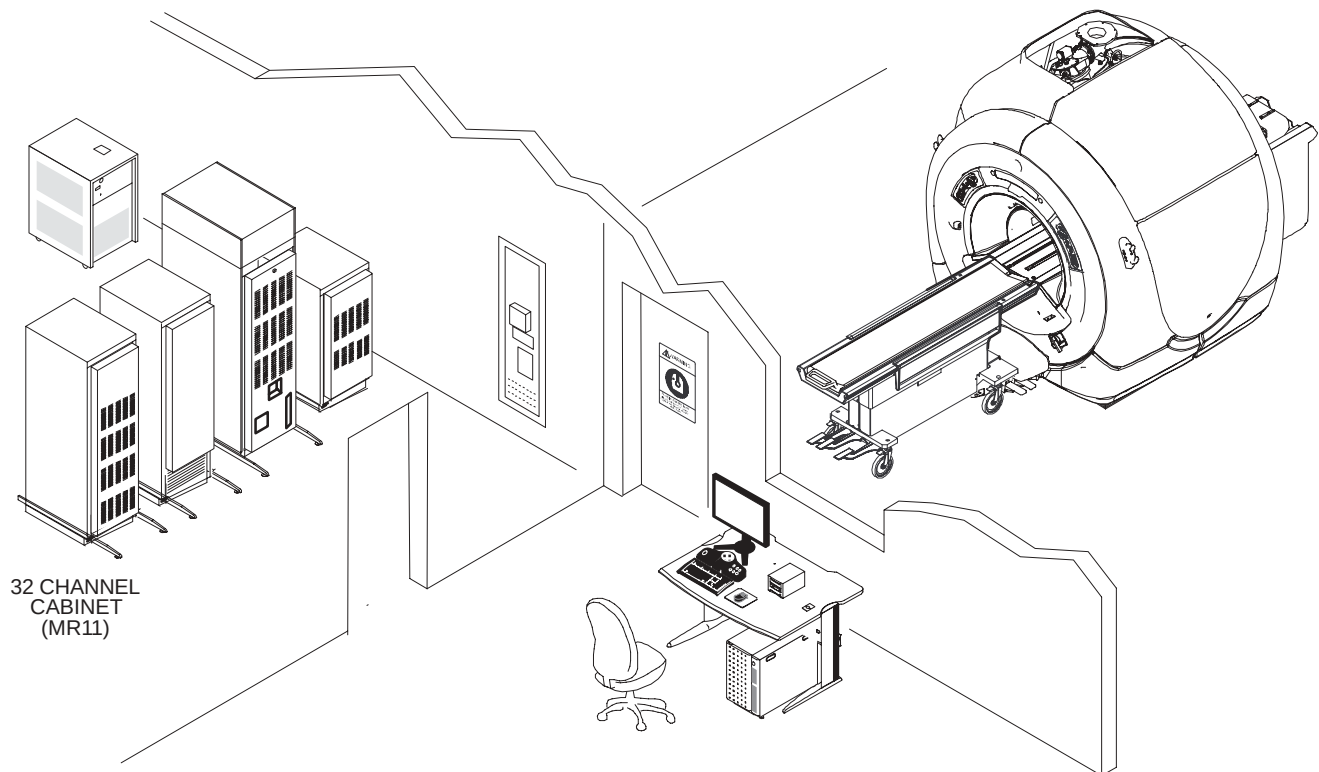
5337894-4 is for a 32 channel system

5337894-3 is for a 16 channel system

Both the 1.5T and 3.0T upgrade will install a new power cable to the HFD cabinet from the 32 Channel cabinet.

6-2 POSITION 32 CHANNEL CABINET

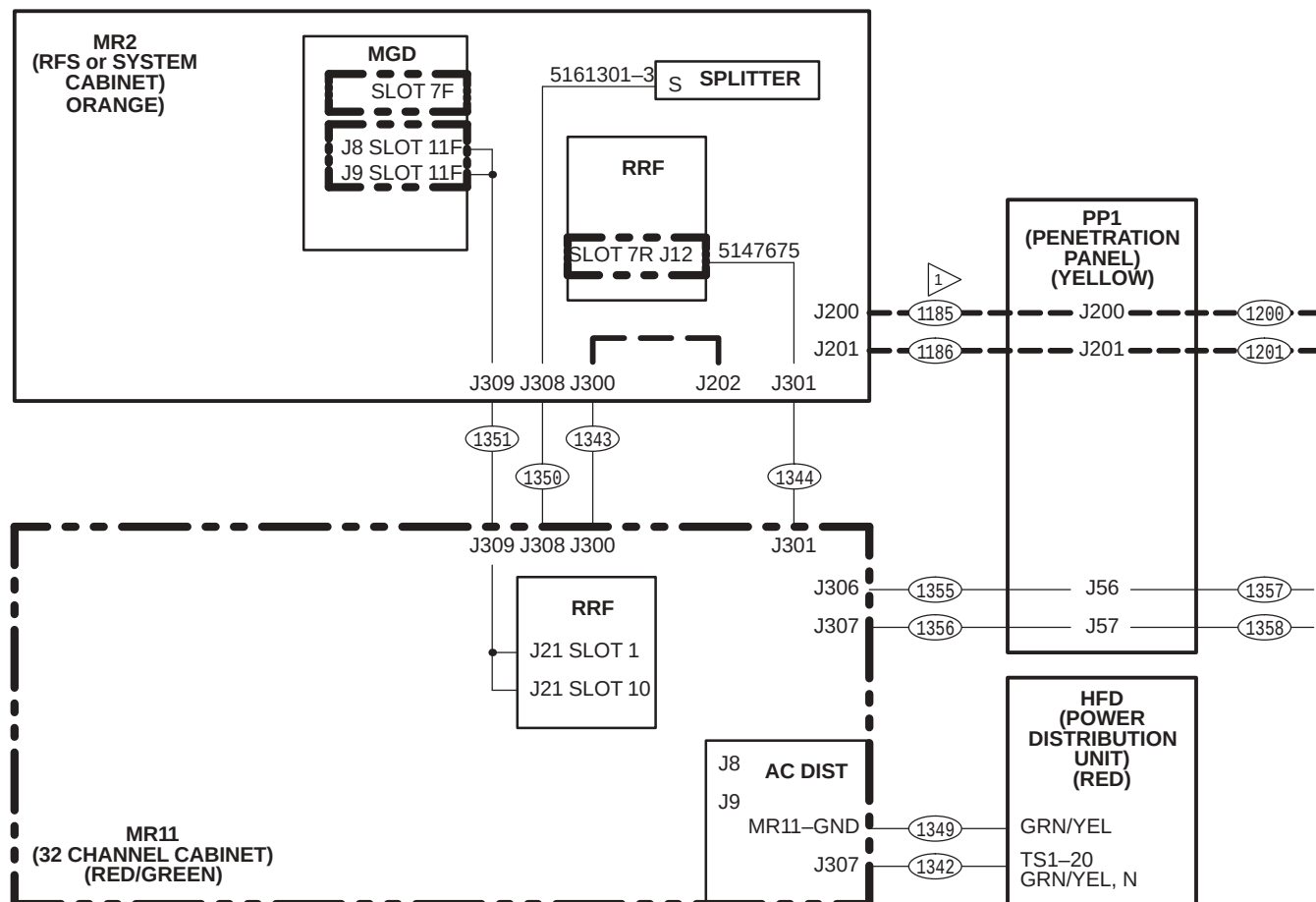
Refer to site upgrade layout drawings for positioning and installation of 32 Channel Cabinet within Equipment Room.



6-3 1.5T UPGRADE CABLE MAP

The cable map below indicates the new cables to be routed and connected along with the new components to be installed in the Equipment Room for a 1.5T system.

Refer to the following procedures for installing these cables and components.



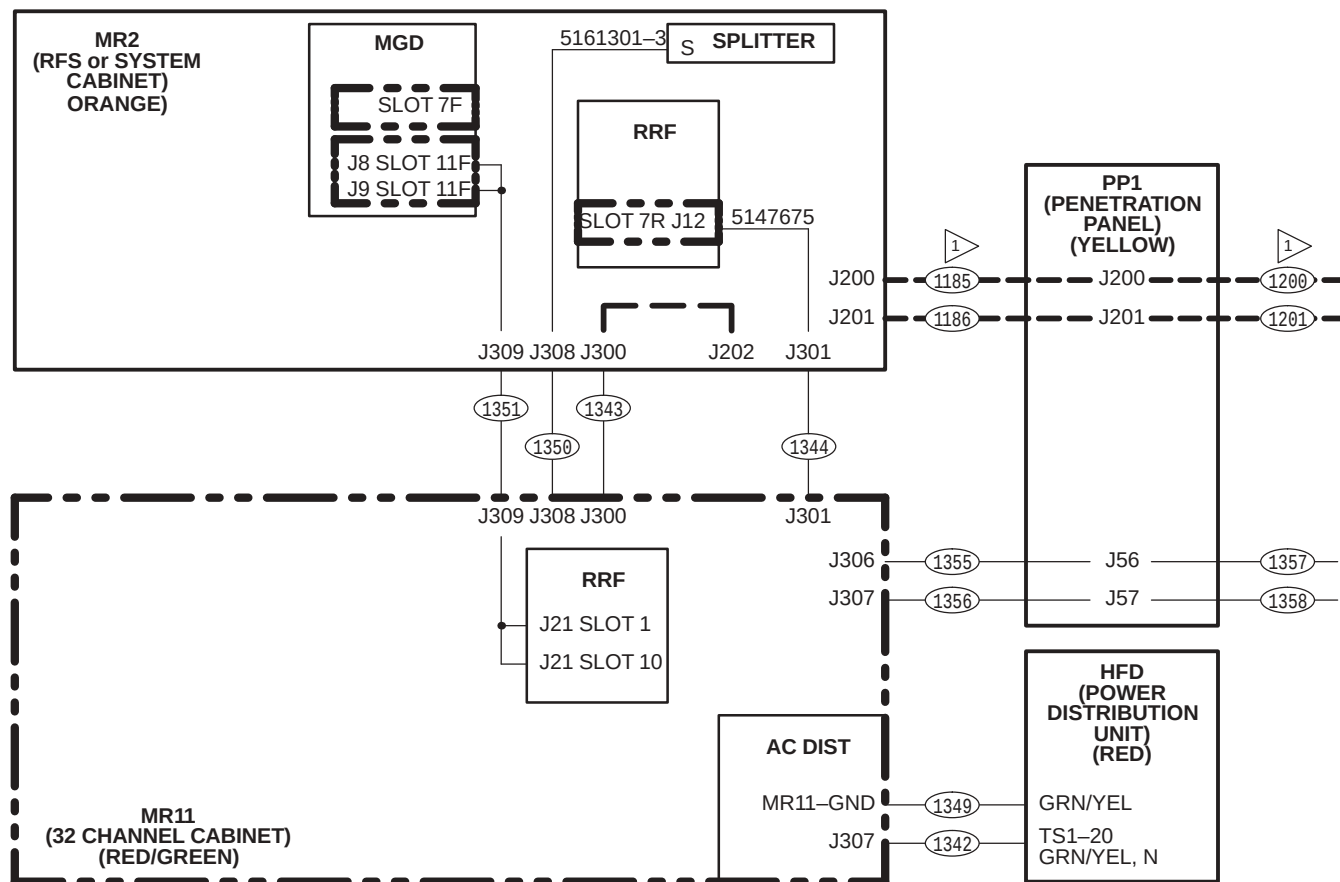
LEGEND:

- EXISTING CABLE
- NEW CABLE
- EXISTING CABLE THAT MAY NEED TO BE REPLACED
- EXISTING CABLE TO BE REROUTED
- UPGRADED CABINET OR MODULE
- NEW CABINET OR MODULE
- REFER TO SECTION 5-4-1 FOR INSTRUCTIONS TO DETERMINE IF RUNS 1185 AND 1186 ARE REQUIRED TO BE REPLACED.

6-4 3.0T UPGRADE CABLE MAP

The cable map below indicates the new cables to be routed and connected along with the new components to be installed in the Equipment Room for a 3.0 system.

Refer to the following procedures for installing these cables and components.



LEGEND:

EXISTING CABLE

NEW CABLE

EXISTING CABLE THAT MAY NEED TO BE REPLACED

EXISTING CABLE TO BE REROUTED

● ● ● ● ●
● ● UPGRADED CABINET OR MODULE

 NEW CABINET OR MODULE

1 REFER TO SECTION 5-4-1 FOR INSTRUCTIONS TO DETERMINE IF RUNS 1185 AND 1186 ARE REQUIRED TO BE REPLACED

6-5 RFS OR SYSTEM CABINET UPGRADE FOR 1.5T AND 3.0T with MGD CHASSIS

Refer to Section 6-6 if cabinet being upgraded has CAM chassis rather than MGD chassis.

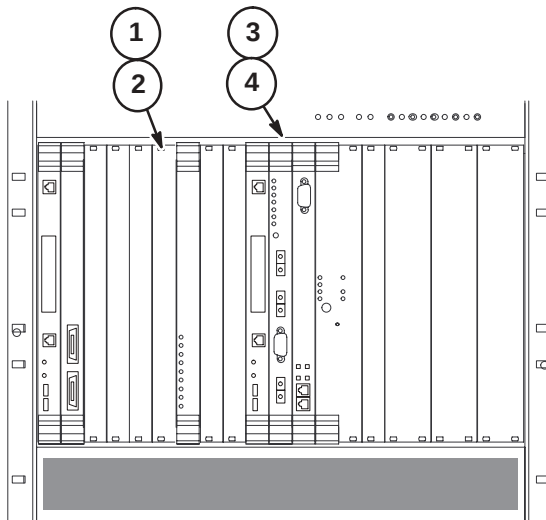


Note: Modules are static sensitive components and need to be handled as such (wrist strap).

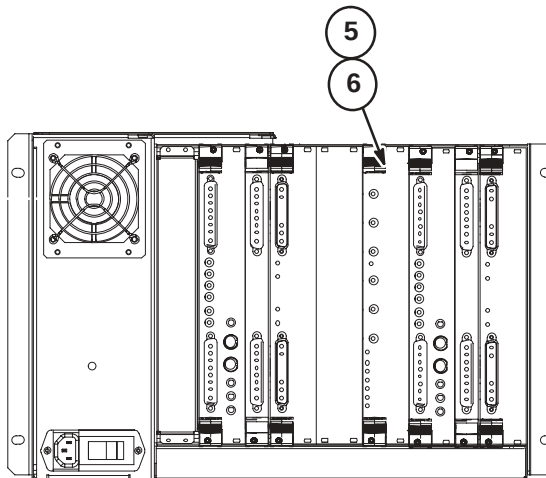
Note: To remove modules – loosen the screws in the ejector pins, then push the top and bottom tabs to eject the boards from the chassis.

6-5-1 MGD AND RRF MODULE UPGRADES

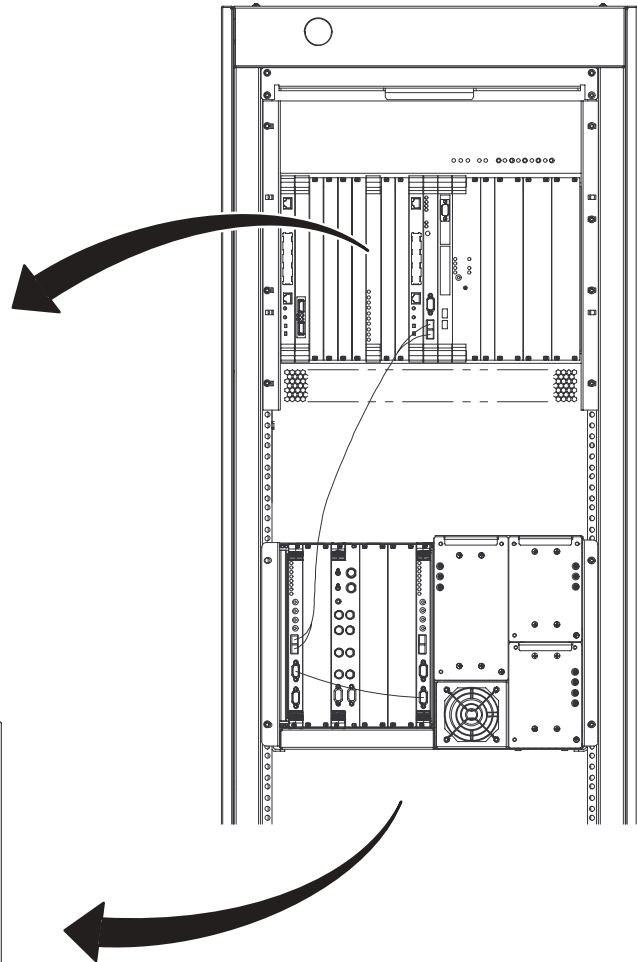
1. Remove blank panel in Slot 6F and discard. Check for blue plastic air baffle in bottom of slot. Remove and discard if present.
2. Install DRF2 Board (2298332) in Slot 6F.
3. Disconnect cables at IRF Board in Slot 11F. Remove existing IRF Board and return for recycling.
4. Install IRF2 Board (2358122) in Slot 11F and reconnect cables.
5. Disconnect cables at Exciter Board in Slots 6/7R. Remove existing Exciter Board and return for recycling.
6. Install Master Exciter 32 Board (5250144) in Slot 6/7R and reconnect cables.



MGD CHASSIS FRONT VIEW

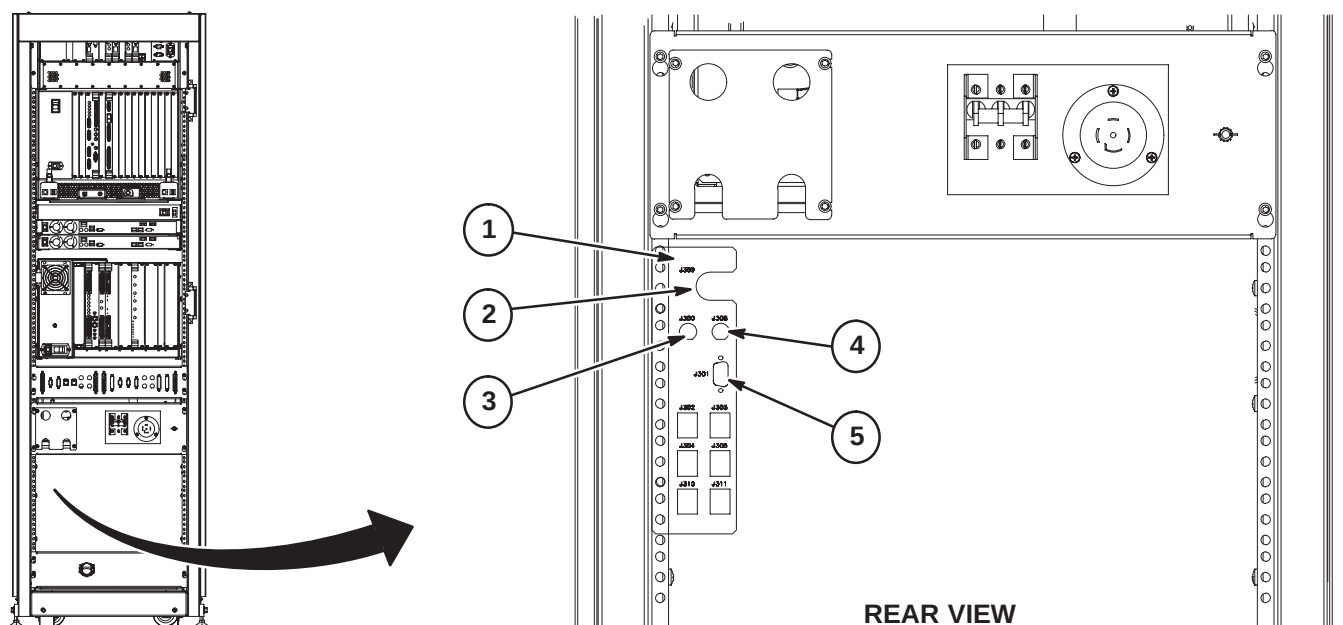


RRF CHASSIS REAR VIEW

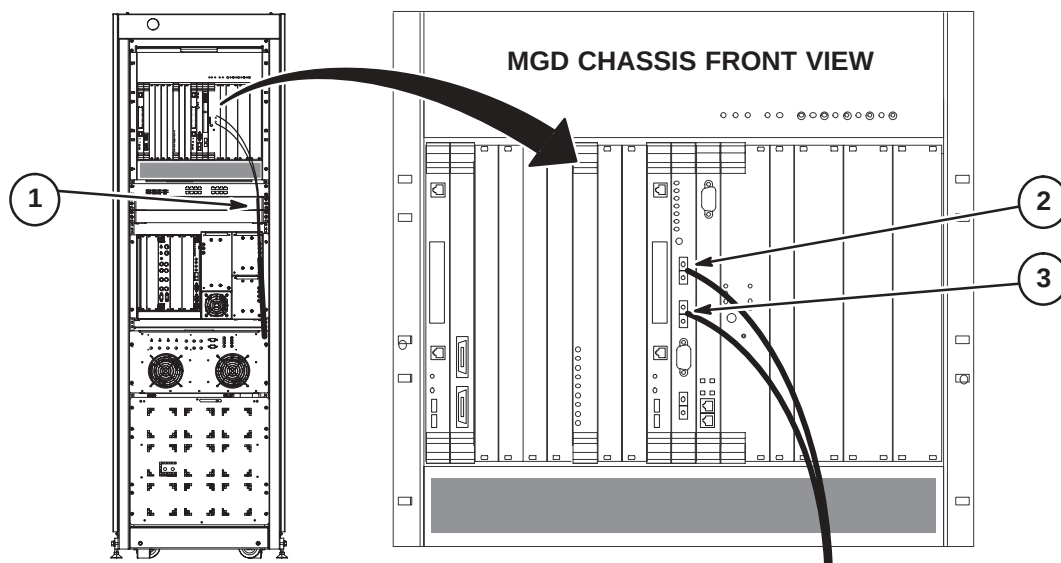


6-5-2 INSTALL RFS OR SYSTEM CABINET I/O PANEL FOR 32 CHANNEL

1. Install 32 Channel System Cabinet I/O Panel (5160806) to back side of vertical channel as shown using two M5 x16mm Pan Head screws (2279742-2).
2. Connect fiber optic Run1351 to J309 and route fiber optic wires under RRF Chassis and up to front of MGD module. Refer to Section 6-5-3 for connection of wires to IRF2 Board.
3. Connect Run 1343 to J300.
4. Connect Run 1350 to J308.
5. Connect Run 1344 to J301.

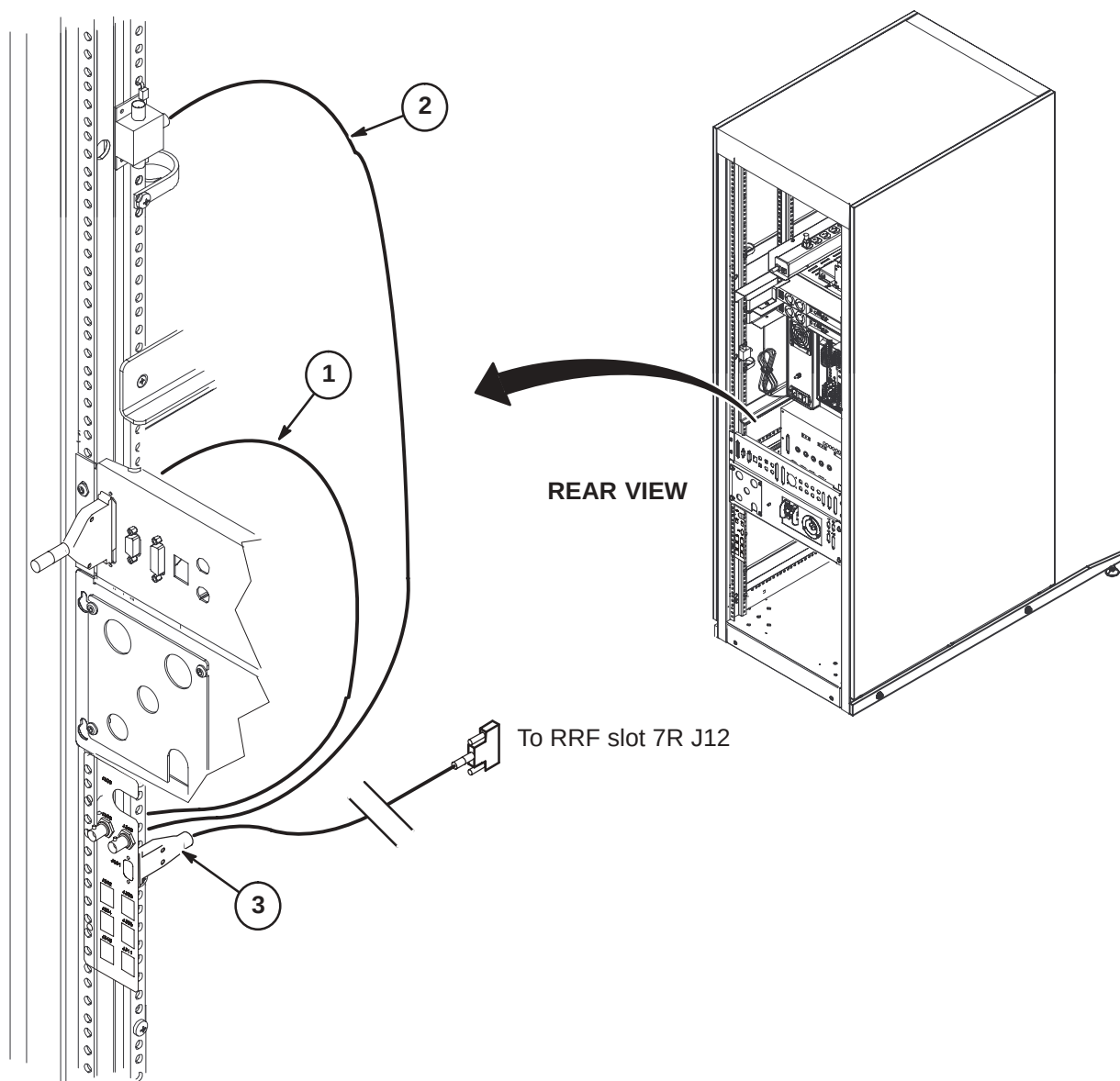
**6-5-3 CONNECT RUN 1351 TO IRF2 BOARD**

1. Route Run 1351 fiberoptic wires with duplex connectors to front of the MGD Chassis to IRF2 Board in slot 11.
2. Connect the BLU/ORN connector to J9.
3. Connect the GRN/BRN connector to J8.



6-5-4 RE-ROUTE EXISTING CABLE AND INSTALL NEW CABLES

1. Disconnect cable connected to "S" on Splitter module and connect to J300 on new I/O Panel.
2. Connect new splitter cable (5161301-3) to "S" on Splitter module and connect other end to J308 on new I/O Panel.
3. Connect new cable (5147675) to J301 on new I/O Panel and route to rear of RRF Module and connect to J12 on new Master Exciter 32 board in slot 6/7R.



6-6 RFS OR SYSTEM CABINET UPGRADE FOR 1.5T AND 3.0T with CAM CHASSIS

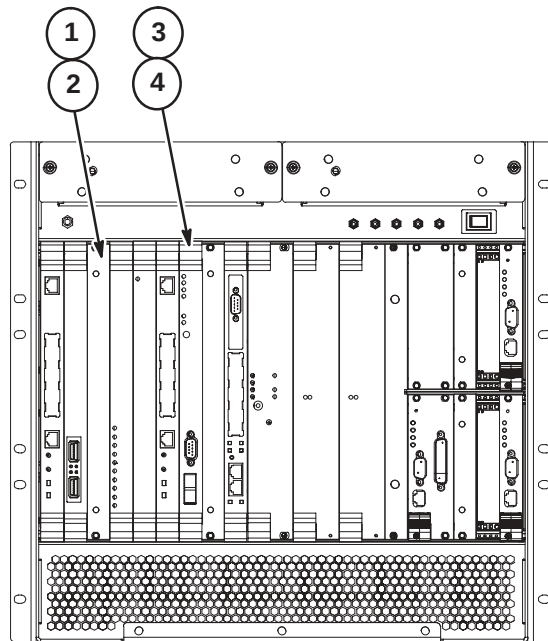


Note: Modules are static sensitive components and need to be handled as such (wrist strap).

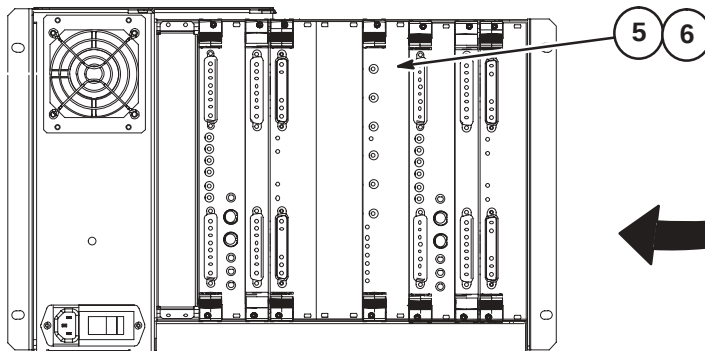
Note: To remove modules – loosen the screws in the ejector pins, then push the top and bottom tabs to eject the boards from the chassis.

6-6-1 CAM AND RRF MODULE UPGRADES

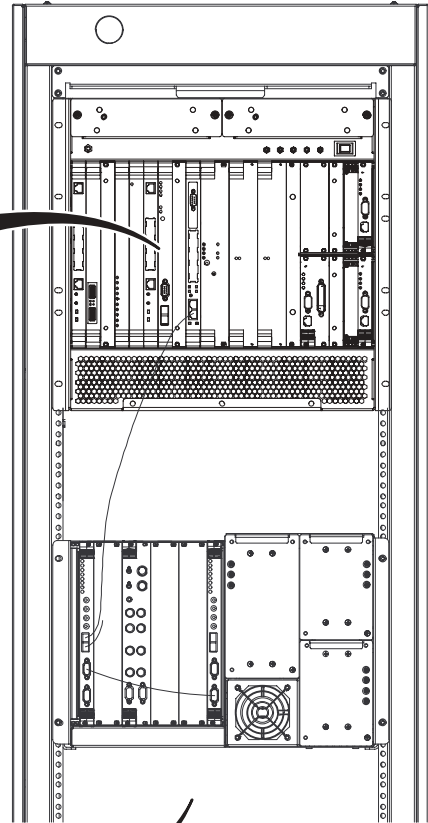
1. Remove blank panel in Slot 3 and discard. Check for blue plastic air baffle in bottom of slot. Remove and discard if present.
2. Install DRF2 Board (2298332) in Slot 3.
3. Disconnect cables at IRF2 Board (2358122–2. 16–channel) in Slot 7. Remove existing IRF2 Board and return for recycling.
4. Install IRF2 Board (2358122, 32–channel) in Slot 7 and reconnect cables.
5. Disconnect cables at Exciter Board in Slots 6/7R of RRF Chassis. Remove existing Exciter Board and return for recycling.
6. Install Master Exciter 32 Board (5250144) in Slot 6/7R and reconnect cables.



CAM CHASSIS FRONT VIEW

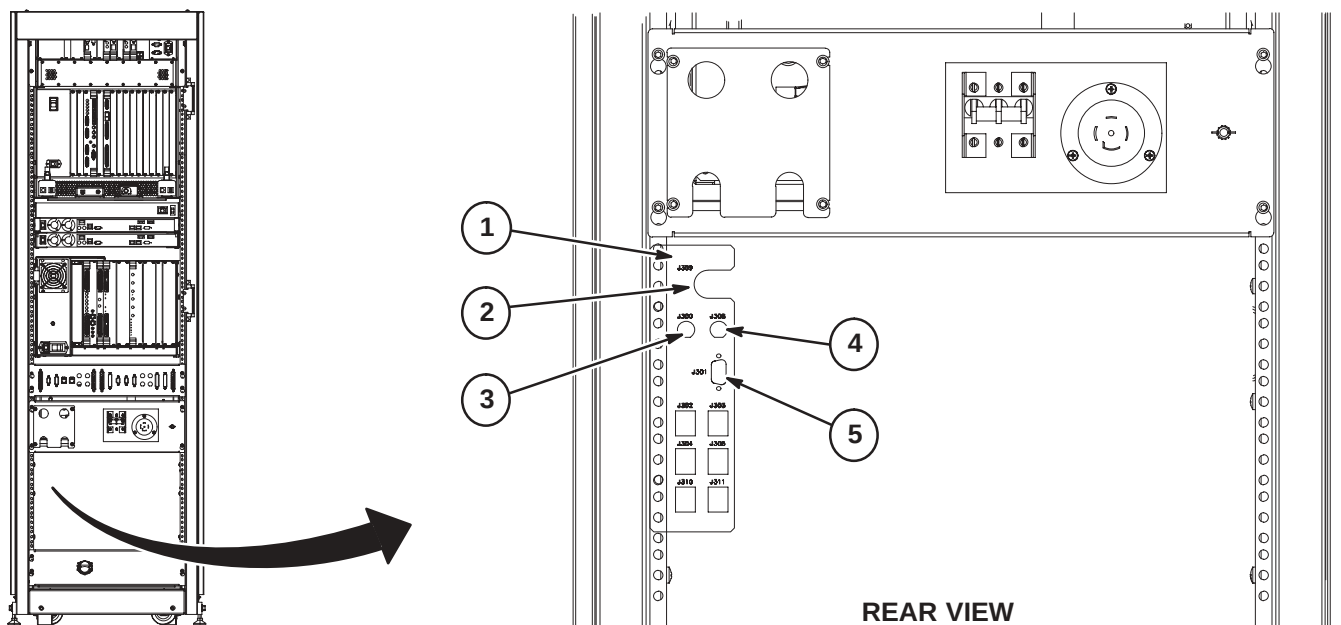


RRF CHASSIS REAR VIEW

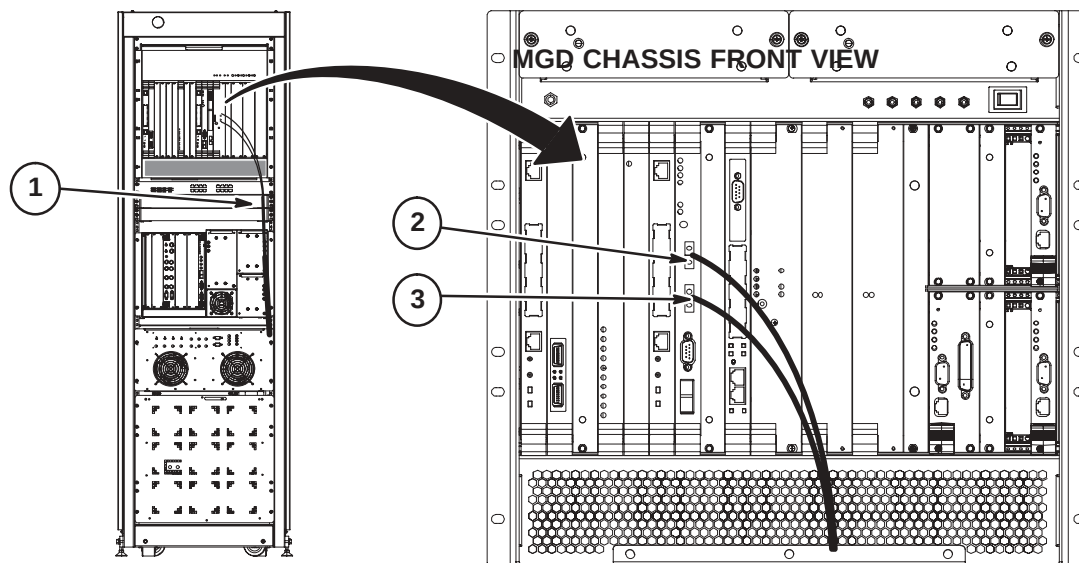


6-6-2 INSTALL RFS OR SYSTEM CABINET I/O PANEL FOR 32 CHANNEL

1. Install 32 Channel System Cabinet I/O Panel (5160806) to back side of vertical channel as shown using two M5 x16mm Pan Head screws (2279742-2).
2. Connect fiber optic Run1351 to J309 and route fiber optic wires under RRF Chassis and up to front of CAM module. Refer to Section 6-6-3 for connection of wires to IRF2 Board.
3. Connect Run 1343 to J300.
4. Connect Run 1350 to J308.
5. Connect Run 1344 to J301.

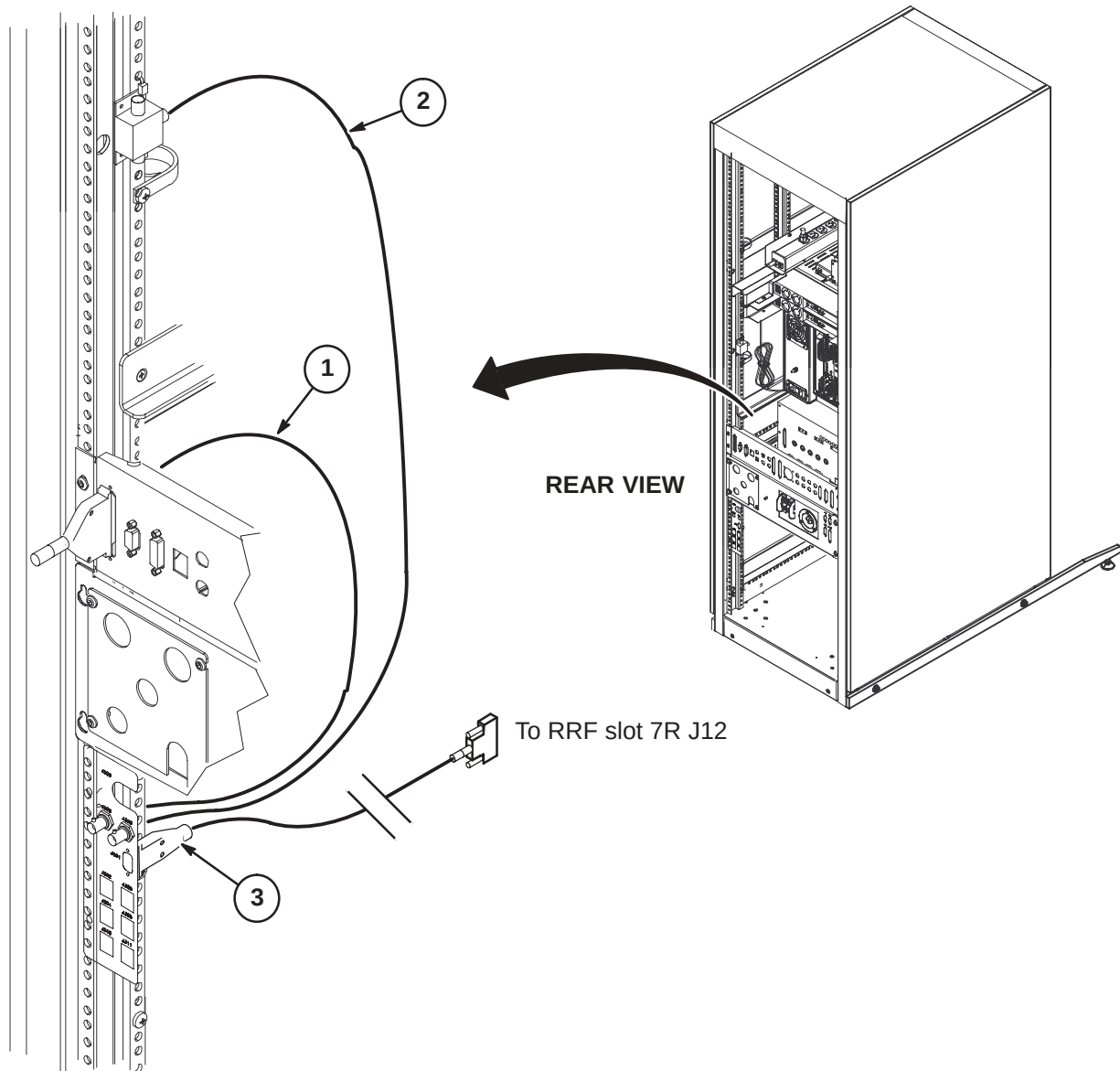
**6-6-3 CONNECT RUN 1351 TO IRF2 BOARD**

1. Route Run 1351 fiberoptic wires with duplex connectors to front of the CAM Chassis to IRF2 Board in slot 7.
2. Connect the BLU/ORN connector to J9.
3. Connect the GRN/BRN connector to J8.



6-6-4 RE-ROUTE EXISTING CABLE AND INSTALL NEW CABLES

1. Disconnect cable connected to "S" on Splitter module and connect to J300 on new I/O Panel.
2. Connect new splitter cable (5161301-3) to "S" on Splitter module and connect other end to J308 on new I/O Panel.
3. Connect new cable (5147675) to J301 on new I/O Panel and route to rear of RRF Module and connect to J12 on new Master Exciter 32 board in slot 6/7R.

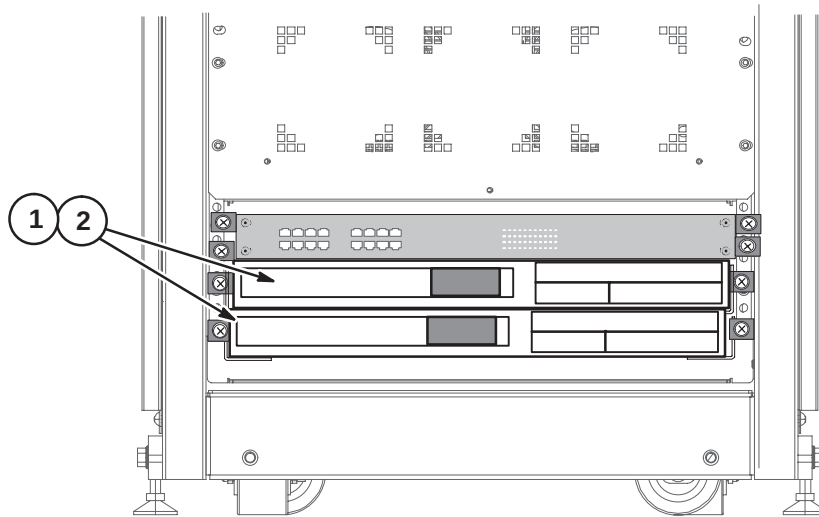


6-7 INSTALL ICN MODULES

6-7-1 ICN UPGRADE FOR CABINET CONTAINING 5127452 ICN's

1. Detach all cables from back of existing installed ICNs.
2. Remove ICN(s) by unfastening (4) M5 screws per ICN and sliding them out. Return for recycling.

NOTE: All existing ICN configurations will upgrade to one ICN, part number 5337894-4.



3. Put 2 (5306485) brackets on the ICN as shown. No additional hardware is needed.

3 On the left side, the bracket is installed downward and snapped into place.

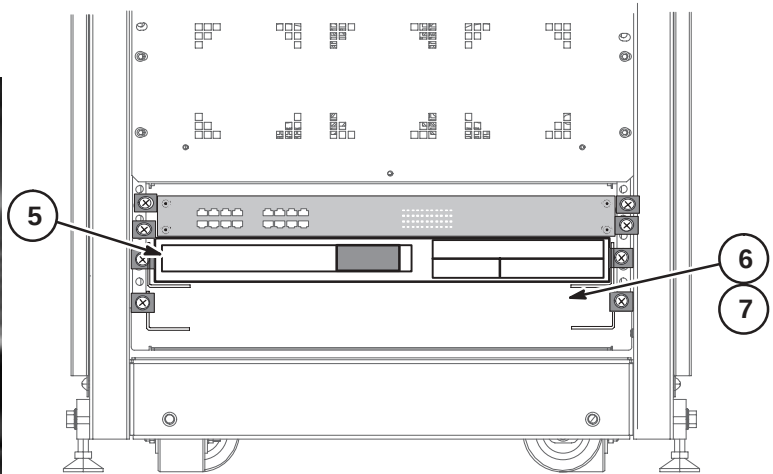


On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.

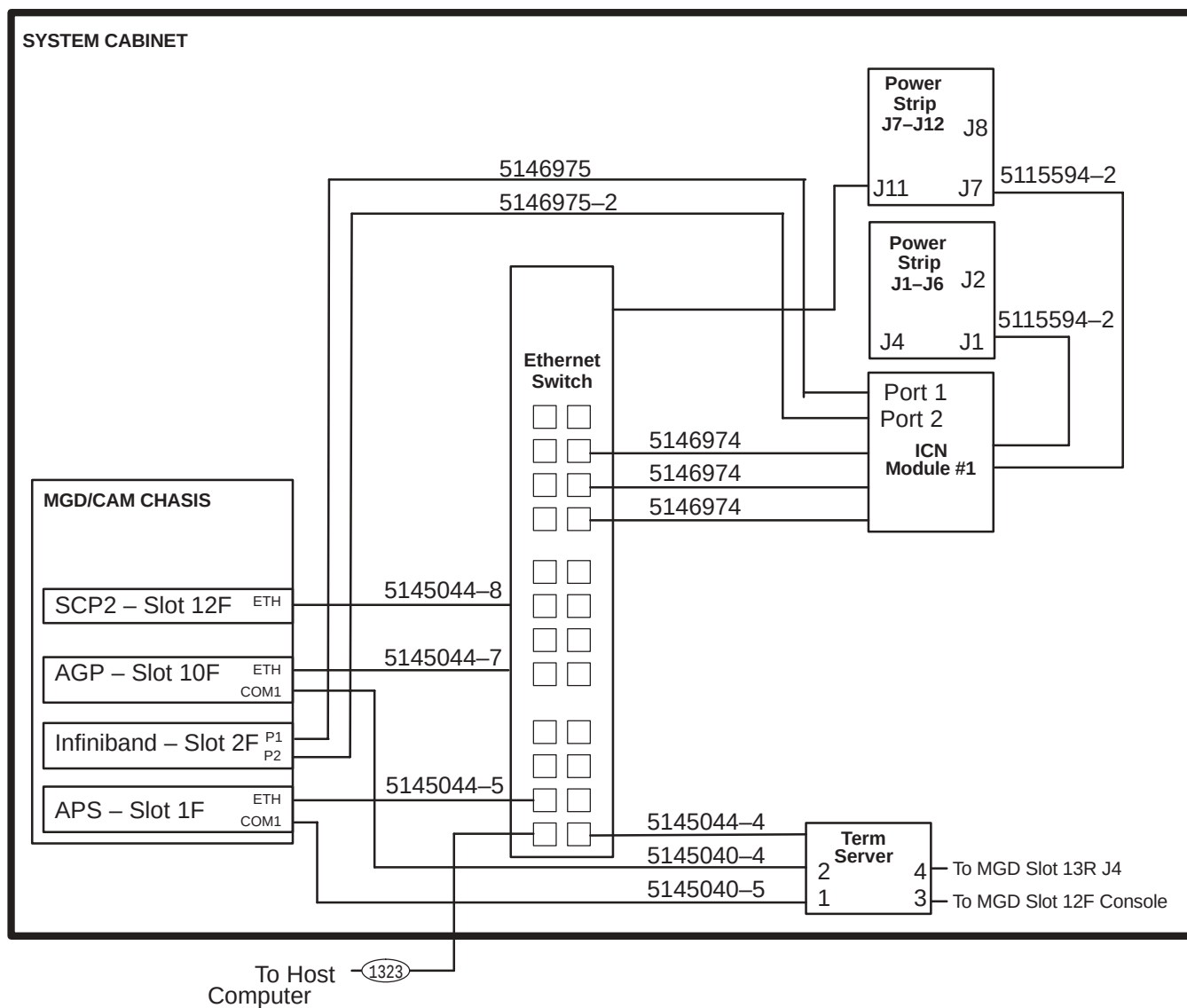


6-7-1 ICN UPGRADE FOR CABINET CONTAINING 5127452 ICN's (Continued)

4. **NOTE:** If the system had DC ICN's, remove the rear bracket as shown below
5. Place then new ICN module (5337894-4) on an open spot where previous ICN was located.
6. Position clip nut over vertical rail hole that aligns with middle hole on each side of the ICN.
7. Secure module in place by tightening the PEM screws on the ICN brackets.

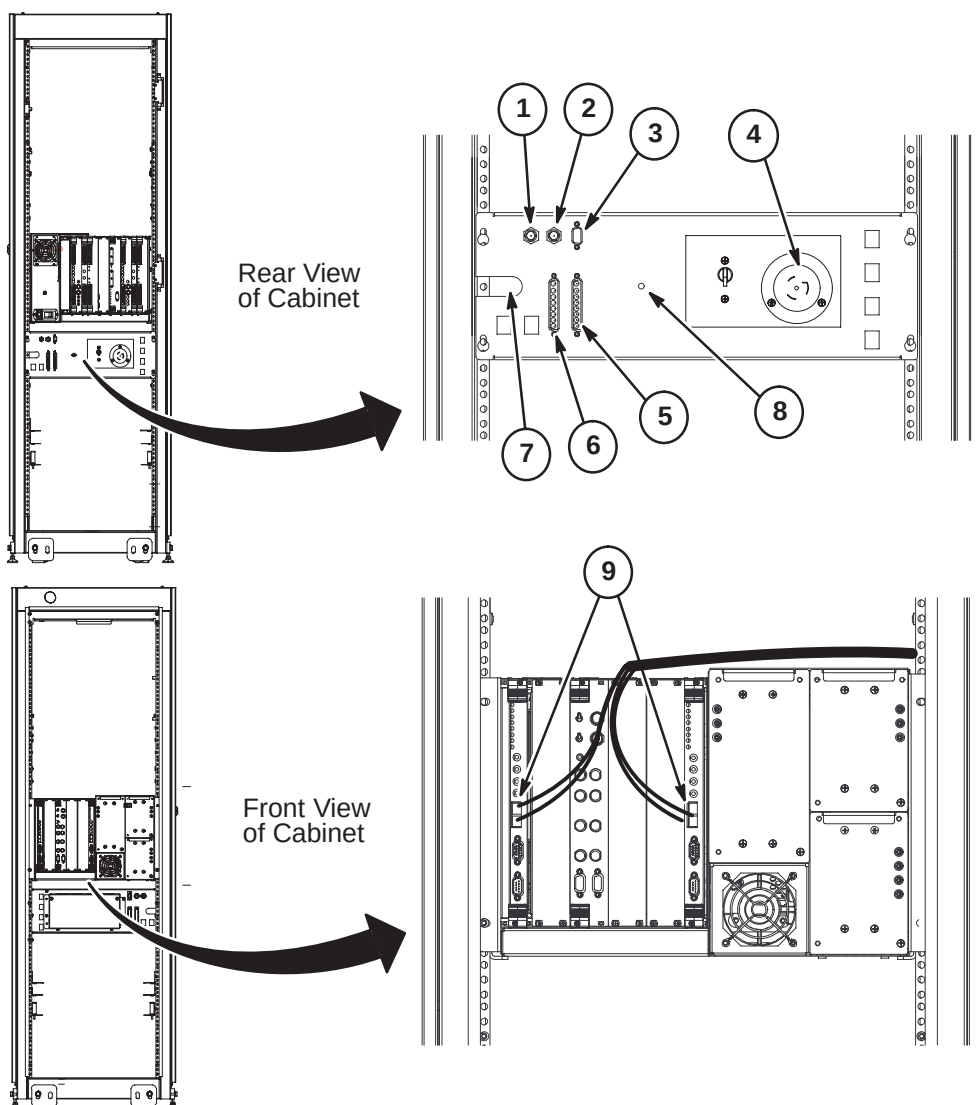


6-7-2 ROUTE AND CONNECT CABLES AT REAR OF MODULES AS SHOWN BELOW



6-8 1.5T and 3.0T INTERFACE PANEL CONNECTIONS FOR 32 CHANNEL CABINET

1. Connect Run 1343 (5161299-2) to J300.
2. Connect Run 1350 (5161299-3) to J308.
3. Connect Run 1344 (5147678) to J301
4. Connect power cable Run 1342 (5161297) to J1. Route other end to HFD cabinet.
5. Route Run 1356 (5160986-2 or -4) from Penetration Panel and connect to J307.
6. Route Run 1355 (5160986 or -3) from Penetration Panel and connect to J306.
7. Connect fiber optic cable Run 1351 (5161300) to J309.
8. Connect Run 1349 GND cable to ground stud.
9. Route the ends of the fiber optic wires along the side of the cabinet, over the top of the RRF Module, and connect to the BLU/ORN connector to J21 on RRF board in slot 1F and the GRN/BRN connector to J21 on RRF board in slot 10F.

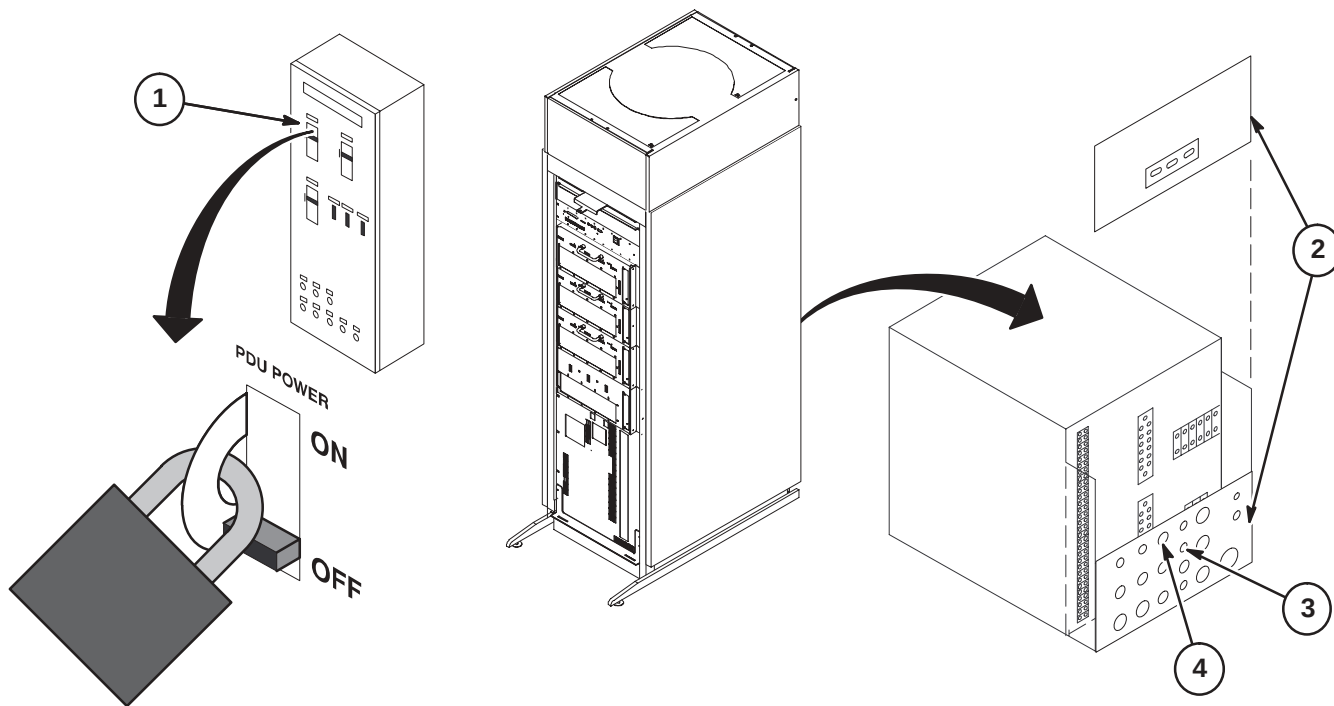


6-9 INSTALL POWER CORD IN HFD CABINET

MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE CONNECTING WIRES.

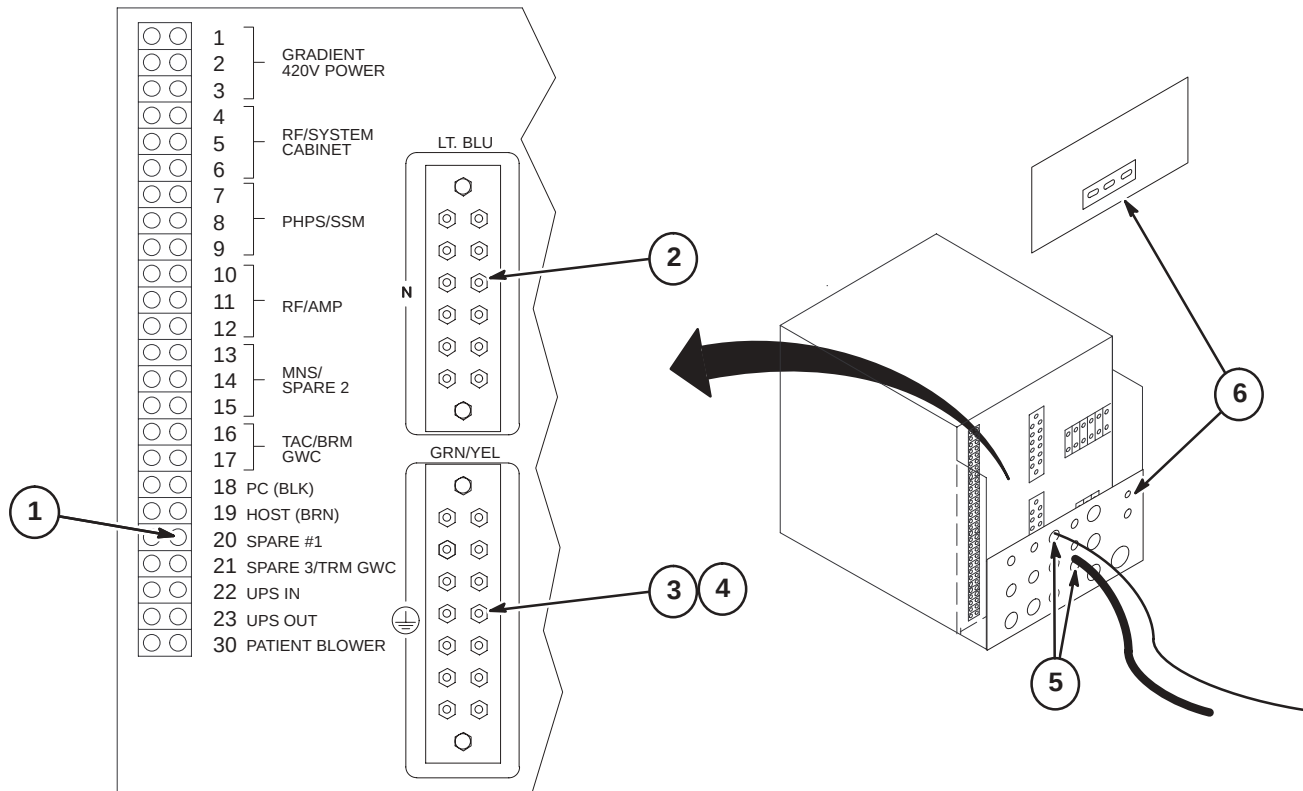
6-9-1 ACCESS POWER STRIP IN PDU MODULE

1. Make sure power is OFF, locked, and tagged at Main Disconnect panel per directions in Section 3-1, before removing covers on PDU module.
2. Remove rear PDU module covers.
3. Select vacant port in interface panel and route power cable Run 1342 (5161297) thru port.
4. Route ground wire cable Run 1349 (5248205) thru the same port as Run 037 (ground for MR2).



6-9-2 CONNECT POWER CABLE TO PDU MODULE

1. Connect BRN wire to Terminal Strip 20 SPARE #1 (Run 1342).
2. Connect BLU wire to Neutral (Run 1342).
3. Connect GRN/YEL wire to GND (Run 1342).
4. Connect Run 1349 to GRN/YEL.
5. Move interface panel into place and tighten clamp around Run 1342 and Run 1349.
6. Install and secure all covers on PDU Module.



SECTION 7 – COMPLETION

7-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 7-2, 7-3, and 7-4.

7-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 7-3. Recycling Operation will route them to appropriate location.

7-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-6. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

7-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers (If required)
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1-8.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 14.x has been made before returning system to the users. Turn site over to applications who will instruct users.

7-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
4. On the PDU:
 - A. Move the mMain Input Circuit Breaker operating handle to OFF position (if in tripped state).
 - B. Move to ON (vertical) position.
 - C. Press the EMO Reset button.
5. Except for the HFD Gradient power modules, switch all cabinet circuit breakers to ON. The HFD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the GOC.